Frequentist and Bayesian Analysis of the Matter–Antimatter Asymmetry in $B^{\pm} \to K^{\pm}K^{+}K^{-}$ Decays at LHCb

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We measure the raw charge asymmetry $A_{\rm raw} = [N(B^-) - N(B^+)]/[N(B^-) + N(B^+)]$ of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays in the 2011 LHCb Open Data record and compare two inference frameworks on the same data and the same question: an extended unbinned maximum-likelihood fit with profile likelihood and a Bayesian analysis sampled with Markov chain Monte Carlo (MCMC). After a common preselection (19,731 candidates in the fit window), a common mass model gives $A_{\rm raw} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst})$ in the frequentist fit and $-0.0443 \pm 0.0085 \pm 0.0070$ in the Bayesian one, 5.2 standard deviations from zero (Bayes factor $\approx 9\times10^3$ against no asymmetry). The two methods agree in value, uncertainty and coverage, which was verified on toy-MC experiments; the dominant systematic uncertainty is the particle-identification requirement. The asymmetry is not uniform over the Dalitz plane: in the region of the published analysis of the same data we find $A_{\rm raw} = -0.234 \pm 0.025$, reproducing -0.239 ± 0.020. A machine-learning selection trained on weak labels, described in Appendix I, leaves $A_{\rm raw}$ unchanged within its statistical spread. $A_{\rm raw}$ is not corrected for production and detection asymmetries and is therefore not a measurement of A_{CP}.

I. INTRODUCTION

Charge-parity (CP) violation – the small observed asymmetry between the behavior of matter and antimatter – is one of the necessary ingredients for the matter-dominated universe we observe, yet the CP violation built into the Standard Model is far too small to account for it. Studying CP violation in the decays of B mesons, where it is comparatively large and experimentally accessible, is one of the main ways to test the Standard Model's Cabibbo-Kobayashi-Maskawa (CKM) description of the effect and search for physics beyond it. Charmless three-body decays such as $B^{\pm} \to h^{\pm}h^{+}h^{-}$ ($h = K, \pi$) are especially interesting in this respect: unlike a two-body decay, the three-body phase space (the Dalitz plot) allows the asymmetry to be studied not only as a single global number, but also *locally*, since interference between intermediate resonances can concentrate CP-violating effects in specific kinematic regions rather than spreading them uniformly [1, 2].

The quantity measured in this work is the raw asymmetry between the signal yields of the two charges,

$$A_{\rm raw} = \frac{N(B^-) - N(B^+)}{N(B^-) + N(B^+)}. \tag{1}$$

It differs from the physical asymmetry A_{CP} by the production asymmetry A_P of $B^{\pm}$ mesons in pp collisions and by the detection asymmetry A_D of the final state; to first order in these small quantities,

$$A_{CP} \approx A_{\rm raw} - A_P - A_D, \tag{2}$$

where A_D arises mainly from the different interaction cross sections of K^+ and K^- with the detector material and depends on the kaon momenta [1]. Neither correction is applied here, so $A_{\rm raw}$ is not a measurement of A_{CP}. The corrections applied in the published analysis are discussed in Sec. IX B.

In this work we use the LHCb Open Data record for $B^{\pm} \to K^{\pm}K^{+}K^{-}$ [3, 4], a subset of the 2011 data on which the first evidence of an inclusive CP asymmetry in this decay was published [5], to ask: *to what extent do frequentist likelihood fits and Bayesian inference produce consistent, well-controlled estimates of the raw asymmetry $A_{\rm raw}$?* The aim is a methodological comparison of the two frameworks on the same data and, by construction, on the same likelihood, so that any difference between the two results comes from the inference and not from the model; it is not a competitive measurement of A_{CP}, which cannot be obtained from this record because it lacks the control channel for A_P and A_D (Appendix J.5); the other three channels of the record need a different selection and a model of peaking backgrounds and are not analysed. Secs. II–III describe the data and the common model, Secs. IV–V the two fits, and Secs. VI–IX the systematic study, the validation, the Dalitz plane and the comparison; a machine-learning selection is in Appendix I. The main results are in Table I.

II. DATA

A. Dataset, Preselection and Mass

In short: 8,556,118 *candidates are reduced to* 23,895 *by a kaon-identification and muon-veto requirement common to the analyses; the fit window* $[5100, 5600]$ *MeV contains* 19,731 *candidates (*10,292 B^+, 9439 B^-*).* The dataset is the CERN Open Data record 4900 [3]: $B^{\pm} \to h^{\pm}h^{+}h^{-}$ candidates collected by LHCb in 2011 at $\sqrt{s} = 7\,\text{TeV}$, already through trigger and offline selection, distributed as ROOT `TTree`s with the most relevant branches and

Method	Sample	$A_{\rm raw}$	Statistical	Systematic	Section
Counting (no fit)	fit window, 19,731	-0.0432	0.0071	–	II B
Frequentist fit	fit window, 19,731	-0.0441	0.0085	0.0072	IV, VI
Bayesian fit	fit window, 19,731	-0.0443	0.0085	0.0070	V, VI
Bayesian fit, D^0 veto	18,744	-0.0496	0.0088	–	D.15
Bayesian fit, ML selection	score > 0.5, 15,709	-0.0451	0.0087	–	App. I.9
Frequentist fit, ML selection	score > 0.5, 15,709	-0.0447	0.0086	–	App. I.9
Bayesian fit, Dalitz region of Ref. [5]	2146	-0.234	0.025	–	VIII
Published analysis (same data) [5]	different selection and model	-0.060	0.007	–	–

TABLE I. Summary of the raw asymmetry $A_{\rm raw}$ (not corrected for production and detection asymmetries). The Bayesian value is the posterior median and its standard deviation; the systematic uncertainty is given only for the nominal fits (Sec. VI). The published $A_{\rm raw}$ comes from a different selection and a model with additional background components, so it is not directly comparable with the first rows.

split by magnet polarity, for 8,556,118 candidates. The record also contains a phase-space simulation of $B^\pm \to K^\pm K^+ K^-$ (50,000 decays, no asymmetry), used only to test the local-asymmetry procedure (Sec. VIII).

For each of the three final-state tracks ($H1$, $H2$, $H3$) the analysis reads the momentum components, the particle-identification (PID) likelihoods (`ProbK`, `ProbPi`), a muon flag and the charge. The same preselection is used in all analyses,

$$\texttt{H*_ProbPi} < 0.5 \ \wedge\ \texttt{H*_ProbK} > 0.5 \ \wedge\ \neg\texttt{H*_isMuon},$$

and retains 23,895 of the 8,556,118 candidates (0.28%), the PID requirement being the dominant reduction (Table V of Appendix A).

$H1$ is always the kaon of charge opposite to the B, so the Dalitz variables are $m^2(H1,H2)$ and $m^2(H1,H3)$.

The three final-state tracks are assigned the charged-kaon mass hypothesis ($M_K = 493.677$ MeV) regardless of their PID likelihoods, and their four-momenta are summed to reconstruct the B candidate, $m(KKK) = \sqrt{(\sum_i E_i)^2 - |\sum_i \vec{p}_i|^2}$. The nominal fit window, common to the frequentist and Bayesian analyses, is $[5100, 5600]$ MeV and contains 19,731 candidates (10,292 B^+, 9439 B^-; 11,636 magnet down, 8095 magnet up). Below ∼5150 MeV the background rises, consistent with a partially reconstructed contribution that a single exponential cannot describe; the window includes the start of this rise, which is described by a second exponential in the background model (Secs. IV–V). The machine-learning selection (Appendix I) uses the same window.

B. Data Exploration

In short: the spectrum peaks at 5284 *MeV with a similar shape for the two charges and the two magnet polarities, and the counting asymmetry of all candidates is* -0.043 ± 0.007.

Figure I shows the spectrum in the fit window, split by B charge (a) and by magnet polarity (b). The peak at $m \approx 5284$ MeV over a falling combinatorial background has the same shape for the two charges and for the two polarities, which is what allows a mass and a resolution common to B^+ and B^- in the model (Sec. III). The asymmetry itself is not visible by eye: the counting asymmetry of all candidates, signal and background together, is $A_{\rm count} = (N^- - N^+)/(N^- + N^+) = -0.043 \pm 0.007$ (binomial uncertainty), and the fits of Secs. IV and V refine it by separating signal and background.

Further material is in Appendix A.

III. MASS MODEL AND LIKELIHOOD

In short: a Gaussian signal and a double-exponential background for each charge, with the raw asymmetry A as a parameter of the fit; both inference methods use the same likelihood. The mass spectrum of each charge sample is described by a signal-plus-background mixture on the nominal fit window $[5100, 5600]$ MeV (Sec. II A),

$$f^\pm(m) = N_s^\pm\, S(m;\mu,\sigma) + N_b^\pm\, B^\pm(m), \qquad (3)$$

where S is a Gaussian truncated to the window, with mean μ and width σ shared by the two charges, and the background is a double exponential with a shape of its own for each charge,

$$B^\pm(m) = f_\pm\, E(m;\lambda_1^\pm) + (1-f_\pm)\, E(m;\lambda_2^\pm),$$

with $E(m;\lambda) \propto e^{-\lambda m}$ a truncated exponential normalized on the window. The steep component (λ_1) describes the rise of the background towards the lower edge of the window, the slow one (λ_2) the combinatorial background under and above the peak. The ordering $\lambda_1 > \lambda_2$ removes the label-switching symmetry of the mixture. Signal and background yields are split between the charges as

$$N_s^\pm = \tfrac{N_s}{2}(1 \mp A), \qquad N_b^\pm = \tfrac{N_b}{2}(1 \mp A_b),$$

so that $A = (N_s^- - N_s^+)/(N_s^- + N_s^+)$, the raw asymmetry $A_{\rm raw}$ of Eq. (1), is itself a parameter of the fit, the parameter of interest (POI, all the others being nuisance parameters), and the background asymmetry A_b is free. The parameter vector has twelve components, $\theta = (N_s, N_b, A, A_b, \mu, \sigma, \lambda_1^\pm, \lambda_2^\pm, f_\pm)$.

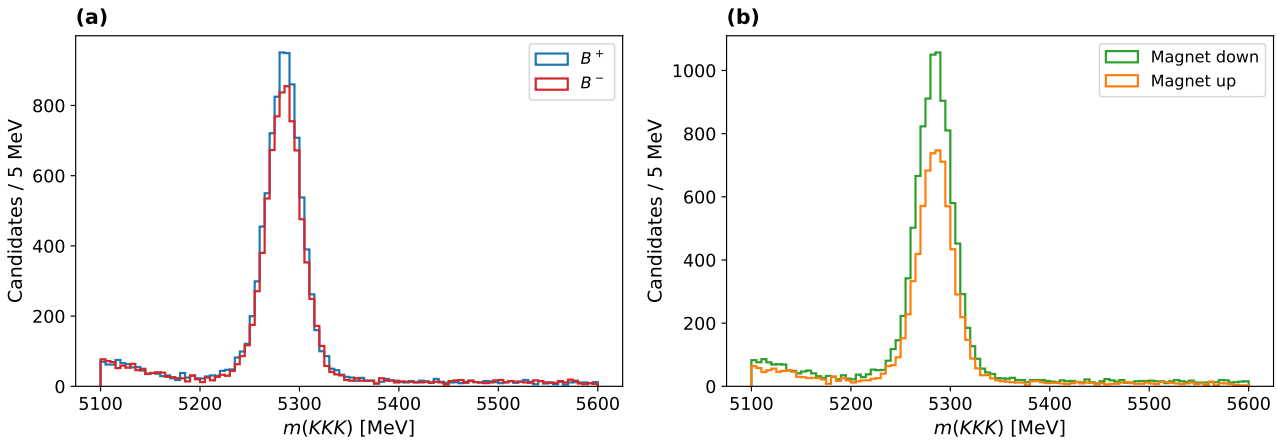


FIG. I. Reconstructed $B \to KKK$ invariant mass in the nominal fit window $[5100, 5600]$ MeV, split by (a) B charge and (b) magnet polarity. Both splits show a consistent peak at $m \approx 5284$ MeV over a falling combinatorial background, which rises below ∼5150 MeV.

Given the data $D = \{(m_i, q_i)\}_{i=1}^{N}$ (mass and charge of each candidate), the extended unbinned likelihood of the two charge samples is

$$\mathcal{L}(D \mid \theta) \propto e^{-(N_s+N_b)} \prod_{i:\, q_i=+1} \Big[N_s^+ S(m_i) + N_b^+ B^+(m_i)\Big] \times \prod_{i:\, q_i=-1} \Big[N_s^- S(m_i) + N_b^- B^-(m_i)\Big], \quad (4)$$

where $S(m_i) \equiv S(m_i; \mu, \sigma)$. The likelihood is extended (the yields are parameters, hence the Poisson term, Appendix B); it is the common ground of the two mass analyses: Sec. IV maximizes it and Sec. V multiplies it by a prior. Since both use this model unchanged, any difference between their results can only come from the inference.

IV. FREQUENTIST ANALYSIS

The frequentist analysis estimates A_{raw} by maximizing the extended unbinned likelihood of Eq. (4), i.e. the same model and the same window $[5100, 5600]$ MeV as the Bayesian analysis, with `iminuit` [6]. The B^+ and B^- samples were fitted simultaneously with A_{raw} as a fit parameter, and its uncertainty and the test of $A = 0$ were derived from the profile likelihood, the other parameters being treated as nuisance parameters (Secs. IV A–IV C).

A. Mass Model and Fit Strategy

The frequentist fit uses the same physical mass model as the Bayesian analysis (Sec. III, Eq. (3)) on the same window $[5100, 5600]$ MeV: a Gaussian signal, with mean μ and width σ shared by the two charges, and a double-exponential background with a shape and a yield of its own for each charge. The twelve parameters are the same as in the Bayesian fit, the two background yields being left free (equivalent to the pair N_b, A_b). They were estimated by maximizing the extended unbinned likelihood of Eq. (4) through `iminuit.cost.ExtendedUnbinnedNLL`: the extended form is needed because the signal and background yields are fit parameters. The minimization started from a background description obtained independently of the signal, with an unbinned fit of the sidebands (the fit window without the region $[5200, 5350]$ MeV) for each charge. The fit gives $N_s = 15{,}327 \pm 133$ signal and $N_b = 4{,}403 \pm 82$ background candidates, to be compared with the Bayesian posterior medians $15{,}328 \pm 133$ and 4404 ± 82 (Sec. V C).

The shapes were not tuned on the frequentist fit: the nominal model is the Bayesian one, and the alternatives are systematic variants (Sec. VI). The nominal fit has $\chi^2/\text{ndof} = 160/88$ ($p = 4.8 \cdot 10^{-6}$) in 2×50 bins of 10 MeV. The double exponential is required by the data: the background rises towards the lower edge of the window (Sec. II A), and a single exponential is rejected ($\chi^2/\text{ndof} = 294/92$ ($p = 5.7 \cdot 10^{-23}$)), so for the systematic study only the double exponential is considered.

B. Separate vs. Simultaneous Fit

The asymmetry was estimated in two ways. In the *simultaneous* fit the two charge samples share the signal parametrization $N_s^+ = N_s(1 - A)/2$, $N_s^- = N_s(1 + A)/2$ (the same one used on the Bayesian side, Sec. III) and the signal mean and width, so that A is a parameter of the likelihood; mass, resolution and background-shape parameters are nuisance parameters, profiled out when reporting uncertainty on A. In the *separate* fit each charge was fitted on its own with the same model and no parameter in common (seven free parameters per charge),

and $A_{\rm raw} = (N_s^- - N_s^+)/(N_s^- + N_s^+)$ was computed afterwards from the two signal yields, with the uncertainty propagated from those of the yields, which are independent because the two samples are. The simultaneous fit gives $A_{\rm raw} = -0.0441 \pm 0.0085$ and the separate fits $A_{\rm raw} = -0.0442 \pm 0.0087$.

The two estimates agree, because sharing μ and σ between B^+ and B^-, as expected for a particle and its antiparticle and a symmetric detector, is adequate. Therefore the simultaneous fit, which estimates A directly and gives the profile-likelihood interval of the next subsection, is taken as the nominal one.

C. Profile Likelihood and Confidence Interval

The maximum-likelihood estimate is $\hat{A} = -0.0441$, with a Hesse uncertainty of 0.0085, and the 68% and 95% Confidence Interval, CI, obtained with `MINOS` [7] are the following:

$$\begin{aligned} A_{\rm raw} &= -0.0441, \\ 68\%\ \text{CI}\ &[-0.0526, -0.0356], \\ 95\%\ \text{CI}\ &[-0.0607, -0.0275]. \end{aligned}$$

The uncertainties are statistical only; the systematic uncertainty is given in Sec. VI.

The profile likelihood ratio,

$$q(A) = -2 \log \frac{L(A, \hat{\hat{\theta}})}{L(\hat{A}, \hat{\theta})},$$

where $\hat{\hat{\theta}}$ are the values of the other eleven parameters that maximize the likelihood at fixed A, was evaluated by re-minimizing them at each value of A (the `mnprofile` of `iminuit`).

For the null hypothesis $A = 0$ it gives $q(0) \simeq 27$, i.e. a p-value of 2.05×10^{-7} corresponding to a significance of 5.20σ: the hypothesis of no asymmetry is rejected. This is the frequentist counterpart of the Savage-Dickey test of Sec. V D, where $A = 0$ lies at 5.2 posterior standard deviations.The two quantities do not measure the same thing (the p-value is not the probability of H_0, and the Bayes factor depends on the width of the prior), but they lead to the same conclusion.

The result agrees with the Bayesian one (Sec. V C: $A_{\rm raw} = -0.0443$, 68% CI $[-0.0528, -0.0358]$). This is expected: with flat priors and $\sim 2 \times 10^4$ events the posterior is dominated by the likelihood and nearly Gaussian, so the credible and confidence intervals coincide numerically although their interpretations differ.

D. Coverage of the Wilks Intervals

Wilks' theorem [8] is an asymptotic result, so the coverage of the profile-likelihood interval on A was checked directly on the data with a bootstrap. 1000 replicas of the 19,731 candidates of the fit window were drawn with replacement, which reproduces the statistical fluctuations of the sample, including those of the charge counts, and the nominal fit was repeated on each of them. The reference value of the asymmetry in the replicas is the best-fit value of the original data, $A_{\rm true} = -0.0441$. For each replica the statistic $q(A_{\rm true})$ was computed as the difference of $-2 \log L$ between the fit with A fixed to $A_{\rm true}$ and the fit with A free and the other eleven parameters profiled. The 68% interval of a replica contains $A_{\rm true}$ if and only if $q(A_{\rm true}) \leq 1$; replicas in which a fit did not converge were discarded.

The fraction of replicas with $q(A_{\rm true}) \leq 1$ is $68.4 \pm 1.5\%$ (binomial uncertainty), against the nominal 68.3%: the interval has the nominal coverage (Fig. II). This agrees with the toy-MC check of the Bayesian credible interval (Sec. D.16, $65.0 \pm 3.3\%$ for the 68% interval); the two values differ by less than one standard deviation, and the uncertainty here is smaller because of the larger number of replicas (1000 against 200). The two checks are not equivalent. The Bayesian pseudo-datasets are generated from the fitted model, which makes that check a closure test, while the replicas here are resampled from the data and therefore carry the features of the spectrum that the model does not describe, such as the shape of the signal tails (Sec. D.13).

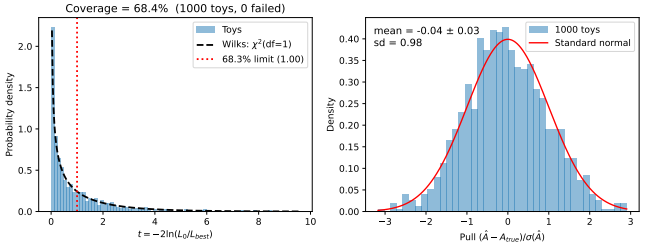


FIG. II. Coverage check, 1000 toys. Left: the statistics $q(A_{\rm true})$ following a $\chi^2_{dof=1}$ distribution as expected from Wilks theorem. Right: pulls $(\hat{A} - A_{\rm true})/\sigma(\hat{A})$, with $\hat{A}$ and $\sigma(\hat{A})$ the fit results of each toy, compared with a standard normal.

E. Systematic Uncertainties and Stability Checks

While the systematic uncertainties study is shared with the Bayesian approach – and discussed in detail in Sec. VI – three specific stability checks were performed within the frequentist one, to evaluate the robustness of the fitted raw asymmetry $A_{\rm raw}$. Each test consists of a two-dimensional grid scan in which two selection or fit parameters are varied simultaneously, performing a simultaneous fit with the nominal double-exponential background model at each point and observing the stability of $A_{\rm raw}$.

1. **Fit Window Scan:** The lower and upper boundaries of the invariant mass fit window, nominally set to $[5100, 5600]$ MeV, were scanned over the ranges $[5000, 5200]$ MeV and $[5350, 6100]$ MeV, respectively, with a 50 MeV step. This test evaluates the sensitivity of A_{raw} to the exponential background parametrization, and is useful to choose the nominal window: too small or too asymmetric fit window produces unstable fit, due to worse background estimation.

2. **Signal Region Scan:** Changing the extremes of the signal region not significant fluctation in A_{raw} are observed, suggesting that the sideband preliminar fit could be avoided.

3. **Particle Identification Cut Scan:** The preselection cuts on the final-state kaon PID likelihoods (`ProbK` and `ProbPi`) were varied simultaneously towards a better kaon purity (larger than 0.5 for kaons, smaller than 0.5 for pions). This influences significantly the A_{raw} estimation, but always within the 68% CI, confirming the cut selection as reasonable.

Across all scan configurations, the maximum observed variation in A_{raw} remains below the statistical uncertainty ($\Delta A_{\text{raw}} < \sigma_{\text{stat}}$), demonstrating the stability of the nominal fit result.

V. BAYESIAN ANALYSIS

The Bayesian analysis takes the likelihood of Sec. III as it is and adds only a prior and a sampler; it changes the question asked of the data, not the model.

A. Priors and Posterior

In short: uniform bounded priors on twelve parameters; with $\sim 2 \times 10^4$ candidates the posterior is dominated by the likelihood. Bayes' theorem combines the likelihood of Eq. (4) (Sec. III) with a prior $\pi(\theta)$ into the posterior

$$p(\theta \mid D) = \frac{\mathcal{L}(D \mid \theta)\, \pi(\theta)}{p(D)}, \tag{5}$$

The priors are uniform, with independent ranges except for the ordering of the background slopes: $N_s, N_b \in (0, 5N)$ with $N = 19{,}731$, $A, A_b \in (-1, 1)$, $\mu \in (5260, 5310)$ MeV, $\sigma \in (3, 40)$ MeV, $\lambda_1^{\pm} \in (10^{-5}, 0.1)$ MeV^{-1}, $\lambda_2^{\pm} \in (10^{-5}, \lambda_1^{\pm})$, $f_{\pm} \in (0, 1)$. The prior density is therefore a constant on the allowed region Θ,

$$\pi(\theta) \propto \mathbf{1}\left[\theta \in \Theta\right], \tag{6}$$

and inside it the posterior is proportional to the likelihood. With $\sim 2 \times 10^4$ events the posterior is dominated by the likelihood, as the prior-sensitivity check of Appendix D.15 confirms for A. The evidence $p(D)$ is a constant: the MCMC samples $\mathcal{L}\,\pi$ without it, and the Savage-Dickey ratio of Sec. V D avoids it as well; the marginal posterior of A is the histogram of the A component of the chain.

B. MCMC Sampling and Convergence Diagnostics

In short: 32 walkers for 25,000 *steps; the chain is at least* 113 *autocorrelation times long for every parameter and independent runs from dispersed starting points agree.* The posterior was sampled with the affine-invariant ensemble sampler of Goodman and Weare [9] as implemented in `emcee` [10] (Appendix D.1): 32 walkers started around the maximum of the posterior, 25,000 steps with the first 1/6 discarded as burn-in (6.7×10^5 samples, acceptance 0.38). The integrated autocorrelation time is $\tau \in [147, 183]$ steps, so the chain is $N/\tau \geq 113$ autocorrelation times long for every parameter, above the threshold of 50 [10] ($\gtrsim 3600$ effectively independent samples). Two runs started uniformly over the whole prior box converge to the same single mode ($A_{\text{raw}} = -0.0446$ and -0.0442; trace plot in Fig. XIII, Appendix D.5).

C. Posterior Results and Credible Intervals

In short: $A_{\text{raw}} = -0.0443$*, 68% credible interval* $[-0.0528, -0.0358]$ *and 95%* $[-0.0608, -0.0276]$*; the posterior is Gaussian.*

Figure III compares the nominal model with the data of the two charges: it follows the spectrum over two orders of magnitude, and the pulls show signal tails heavier than a Gaussian, whose effect on A_{raw} is negligible (Appendix D.13). The marginal posterior on the raw asymmetry gives (median and equal-tailed credible intervals, CI)

$$\begin{gathered} A_{\text{raw}} = -0.0443, \\ 68\%\ \text{CI}\ [-0.0528, -0.0358], \\ 95\%\ \text{CI}\ [-0.0608, -0.0276]. \end{gathered}$$

The posterior is very close to Gaussian (standard deviation 0.0085), so equal-tailed and highest-posterior-density intervals coincide; with uniform priors its mode, -0.0441, is the maximum-likelihood estimate and the result can be compared directly with the frequentist fit (Sec. IV, Appendix C.1). The other physics parameters (Fig. IV) are $N_s = 15{,}328 \pm 133$, $N_b = 4404 \pm 82$, $A_b = -0.040 \pm 0.018$, $\mu = 5284.39 \pm 0.16$ MeV and $\sigma = 18.00 \pm 0.14$ MeV. A is almost uncorrelated with them, which is why profiling (frequentist) and marginalizing (Bayesian) the nuisance parameters give the same result; the largest correlation, -0.17, is with A_b: conditioning on $A_b = 0$ moves A to -0.0475, as the fit with a charge-symmetric background does (-0.0471, Sec. VI).

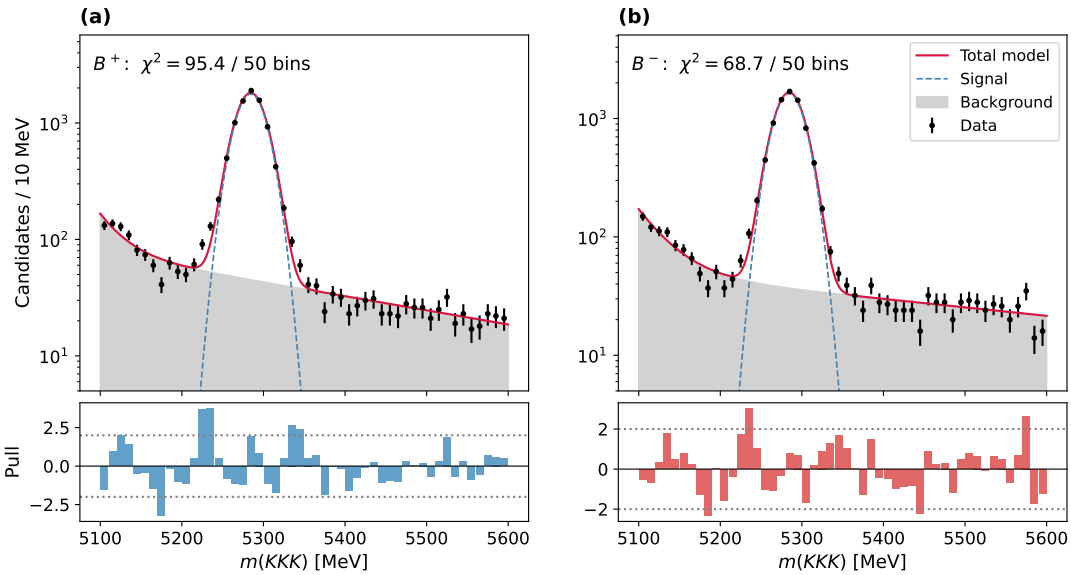


FIG. III. Mass distributions of (a) B^+ and (b) B^- candidates with the nominal model (posterior medians, logarithmic scale): total, signal and background components. Bottom: pulls, $(\text{obs} - \text{exp})/\sqrt{\text{exp}}$.

D. Bayes Factors

In short: decisive evidence against $A = 0$ (Bayes factor $BF_{10} \approx 9 \times 10^3$); the evidence for a background asymmetry is not significant, and depends on the prior width.

The Bayes factor compares two hypotheses through their evidences, $BF_{10} = P(D \mid H_1)/P(D \mid H_0)$, and requires proper priors. For the nested hypothesis H_0: $A = 0$ against H_1: A free, with shared priors on the other parameters, it reduces to the Savage-Dickey ratio [11] (Appendix C.1),

$$BF_{01} = \frac{p(A = 0 \mid D)}{\pi(A = 0)}, \qquad \pi(A = 0) = \tfrac{1}{2}.$$

$A = 0$ lies 5.2 posterior standard deviations from the median, where the chain has essentially no samples, so $p(A = 0 \mid D)$ was taken from the Gaussian approximation of the marginal posterior: $BF_{01} \approx 1.1 \times 10^{-4}$ ($BF_{10} \approx 9 \times 10^3$), decisive evidence against $A = 0$ on Jeffreys' scale [12]. The value scales with the prior width (Ockham factor): with $A \in (-0.3, 0.3)$, BF_{01} drops by a factor 3.3 and the conclusion is unchanged (Fig. V; the Savage-Dickey densities are in Fig. XIV of Appendix D, further checks in Appendix H.3).

The same test on the background asymmetry shows the Ockham factor at work: $A_b = -0.040 \pm 0.018$ is 2.3 standard deviations from zero, yet $BF_{01}(A_b = 0) \approx 3.5$, because the likelihood gain does not compensate for spreading the prior over $(-1, 1)$; unlike for A, the preference changes sign with the prior width (Fig. V, crossing at $W \approx 0.3$). Of the Bayesian results only the Bayes factor depends on the prior; the interval of A does not (Sec. V C). A negative A_b is plausible given the misidentified $B^\pm \to K^\pm\pi^+\pi^-$ and $\pi^\pm K^+K^-$ decays; A_b is kept free in the nominal model and fixing it to zero is one of the systematic variants (Sec. VI).

Further material is in Appendix D. *Model Adequacy: Goodness of Fit and Posterior Predictive Check*: the model does not reach $\chi^2/\text{ndof} \approx 1$ (164.1/88), mainly because the signal tails are heavier than a Gaussian, but A_{raw} changes by only -0.0002 with a Crystal Ball signal.

VI. SYSTEMATIC UNCERTAINTIES AND RESULT

In short: thirteen variants in four groups give a systematic uncertainty of 0.0070 *(Bayesian) and* 0.0072 *(frequentist), dominated by the particle-identification requirement; the final result is* $A_{\text{raw}} = -0.0443 \pm 0.0085\,(stat) \pm 0.0070\,(syst)$. The systematic study asks how much the result depends on choices of the analysis that the data do not fix. The frequentist and Bayesian fits share one study, so that the two uncertainties are comparable: the same thirteen variants of the nominal choices, fitted by both methods on exactly the same candidates, and the same rule to combine them. The variants (listed in Table II) change one choice at a time and fall in four groups: the *window*, the *background* model, the *signal* model and the *preselection* cuts.

The shift $\Delta = A_{\text{raw}}^{\text{var}} - A_{\text{raw}}^{\text{nom}}$ of a variant is not necessarily a systematic effect: when a variant changes the sample (window, preselection) the two samples are nested and part of Δ is a statistical fluctuation of expected size $\sigma_B = \sqrt{|\sigma^2_{\text{var}} - \sigma^2_{\text{nom}}|}$ [13], and such a variant contributes only if $|\Delta| > \sigma_B$. A variant that the data reject ($\Delta\chi^2 > 25$ on the same data) is excluded: this removes the single exponential ($\Delta\chi^2 = 134$, Appendix D.13). The

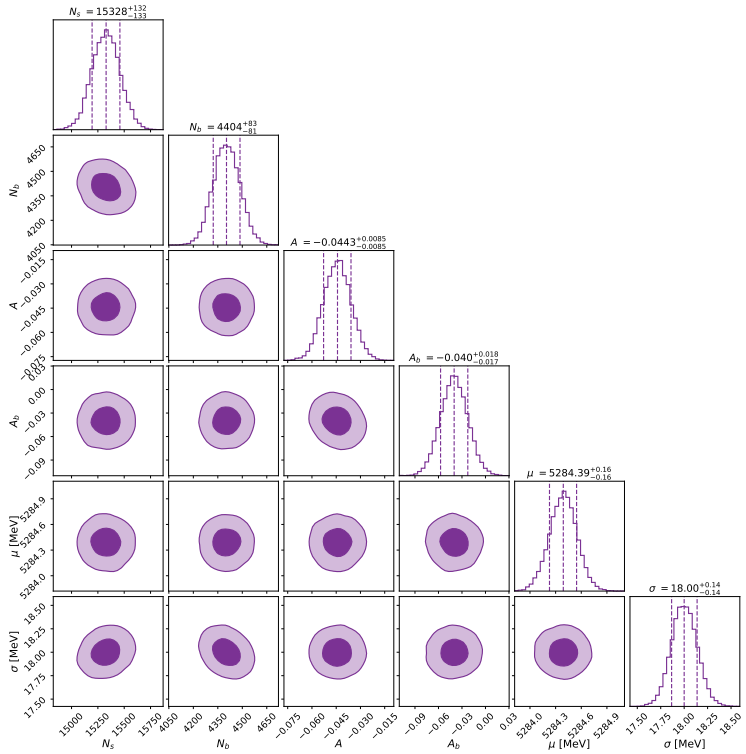


FIG. IV. Posterior of the six physics parameters of the nominal fit (the six background-shape parameters are marginalized). Diagonal: marginal posteriors with the 16th, 50th and 84th percentiles (dashed lines and titles). Off-diagonal: 2D marginals, contours enclosing the regions of the 1σ and 2σ levels of a 2D Gaussian (39% and 86%).

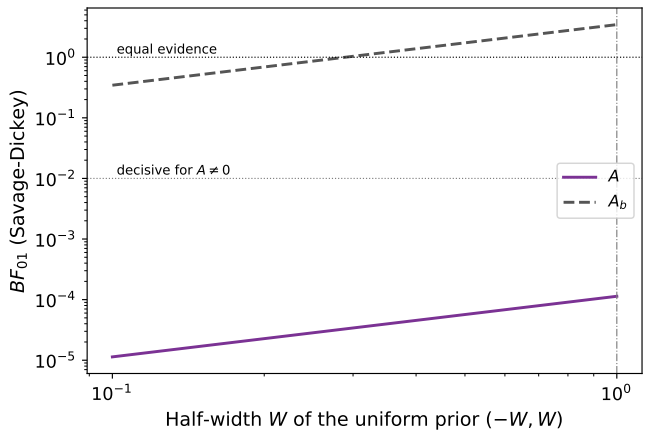


FIG. V. BF_{01} of $A = 0$ and of $A_b = 0$ as a function of the half-width W of the uniform prior $(-W, W)$ (Gaussian approximation of the posterior, Savage-Dickey). The vertical line marks the nominal prior.

largest remaining $|\Delta|$ of each group is its uncertainty, since the variants of a group are alternative descriptions of the same choice, and the groups, being independent choices, are added in quadrature.

The results are in Table II, and all the Bayesian fits in Table XII of Appendix E.1. The two methods give the same shifts to within ∼0.0005 and the same group uncertainties, with totals of 0.0070 (Bayesian) and 0.0072 (frequentist):

$$\text{Bayesian:}$$
$$A_{\text{raw}} = -0.0443 \pm 0.0085\,(\text{stat}) \pm 0.0070\,(\text{syst}),$$
$$\text{frequentist:}$$
$$A_{\text{raw}} = -0.0441 \pm 0.0085\,(\text{stat}) \pm 0.0072\,(\text{syst}).$$

The preselection dominates, through the single variant `ProbK` > 0.6 ($+0.0058$, $1.2\,\sigma_B$). The scan of each cut in steps of 0.05 (Fig. XX of Appendix E) shows that this value is a fluctuation: on the tight side the shifts scatter around zero within $1.3\,\sigma_B$ without a trend, whereas loosening `ProbK` moves A_{raw} monotonically, as expected if misidentified decays not in the mass model enter the sample. The nominal requirement is therefore well placed and the preselection uncertainty conservative. The background group is second, driven by the charge dependence of the background, and the window third.

VII. VALIDATION OF THE ESTIMATOR AND OF THE INTERVALS

In short: toy-MC experiments generated from the fitted model show that the maximum-likelihood estimate of A is unbiased and attains the minimum variance bound, that

Group	Variant	n	Δ Bayes	σ_B Bayes	Δ freq.	σ_B freq.
Window	$[5050, 5600]$	21164	-0.0003	0.0009	-0.0004	0.0005
	$[5150, 5600]$	18566	-0.0001	0.0007	-0.0005	0.0021
	$[5100, 5450]$	19004	$+0.0002$	0.0009	-0.0001	0.0003
	$[5100, 5700]$	20191	$\mathbf{-0.0014}$	0.0010	$\mathbf{-0.0014}$	0.0001
Background	single exponential (rejected)	19731	-0.0037	–	-0.0039	–
	shape shared between charges	19731	$\mathbf{-0.0037}$	–	$\mathbf{-0.0040}$	–
	symmetric, $A_b = 0$	19731	-0.0028	–	-0.0031	–
Signal	Crystal Ball	19731	$\mathbf{-0.0002}$	–	$\mathbf{-0.0006}$	–
	μ, σ per charge	19731	-0.0000	–	-0.0001	–
Preselection	`ProbK` > 0.4	26553	-0.0032	0.0038	-0.0032	0.0037
	`ProbK` > 0.6	13981	$\mathbf{+0.0058}$	0.0048	$\mathbf{+0.0058}$	0.0050
	`ProbPi` < 0.4	17984	$+0.0022$	0.0022	$+0.0020$	0.0023
	`ProbPi` < 0.6	20953	-0.0004	0.0020	-0.0006	0.0017
Group uncertainties: window / background / signal / preselection				0.0014/0.0037/0.0002/0.0058		0.0014/0.0040/0.0006/0.0058
Total (quadrature)				0.0070		0.0072

TABLE II. Systematic study common to the Bayesian and frequentist fits: shift of A_{raw} for each variant and, for the variants that change the sample, the expected statistical spread σ_B of the shift. The largest contributing shift of each group is in bold. The single exponential is rejected by the data and excluded. For the window $[5150, 5600]$ the frequentist uncertainty is scaled from the number of events, since the Hesse matrix of that fit is not accurate.

the Wilks interval has the nominal coverage, and that the test of $A = 0$ has the expected size and a power that is known as a function of the injected asymmetry.

The asymptotic properties of the maximum-likelihood estimator and of the profile-likelihood ratio were also verified on toy MC experiments generated from the fitted model of Sec. III (Poisson yields, masses drawn from the model), as done for the Bayesian interval in Appendix D.16: 1000 experiments at the fitted value $A_{\text{true}} = -0.0441$ and 300 at each of -0.1, -0.02, 0 and $+0.05$, each fitted with the nominal frequentist procedure (1–3% of the fits are not flagged valid by `MIGRAD` and are excluded).

A. Bias, Efficiency and Coverage

In short: no bias larger than 0.0008*, an estimator spread equal to the Hesse uncertainty within* 6%*, and coverage of* 0.70 ± 0.01 *and* 0.95 ± 0.01 *at the fitted value; the test of $A = 0$ has size* 0.051 ± 0.013 *and power* 0.68 *at* $A = -0.02$. Table III summarizes the estimator over the five injected values: the bias of $\hat{A}$ is at most 0.0008 (1.6 standard errors), its spread equals the mean Hesse uncertainty within 6% and the pulls have width 0.94–1.05, so the estimator is unbiased and reaches the minimum variance bound. At the fitted value the coverage of $t(A_{\text{true}}) \leq 1$ is 0.696 ± 0.015 (68.3% level) and 0.954 ± 0.007 (95.4% level); at $A_{\text{true}} = +0.05$ the 68% coverage is 0.614 ± 0.028, 2.5 standard deviations low, not unexpected among five values.

Size and power. Under $A_{\text{true}} = 0$ the test $q(0) > 3.84$ rejects in 0.051 ± 0.013 of the toy experiments, the nominal size, and $q(0)$ is compatible with χ^2_1 (Kolmogorov-Smirnov $p = 0.79$). Figure VI gives the power: 1.00 at the fitted value and 0.68 (0.27 at 3σ) at $A_{\text{true}} = -0.02$,

A_{true}	n	bias	$\text{sd}(\hat{A})/\langle\sigma_{\text{Hesse}}\rangle$	cov. 68%	cov. 95%
-0.1	297	-0.0005 ± 0.0005	0.94	0.71	0.96
-0.0441	985	-0.0001 ± 0.0003	0.97	0.70	0.95
-0.02	290	-0.0008 ± 0.0005	1.02	0.69	0.94
0	293	$+0.0002 \pm 0.0005$	1.01	0.69	0.95
$+0.05$	293	-0.0004 ± 0.0005	1.05	0.61	0.96

TABLE III. Toy-MC experiments at five injected values of the asymmetry: number of valid fits, bias of $\hat{A}$ with its standard error, spread of the estimates relative to the mean Hesse uncertainty (0.0085), and coverage of the intervals $t(A_{\text{true}}) \leq 1$ and ≤ 3.84.

so with 19,731 candidates an asymmetry below ~ 0.02 is not resolved. Together, the estimator is unbiased and efficient, the interval has its nominal coverage and the test its nominal size; the 5.2σ of the data is an extrapolation of the χ^2_1 tail, which the 293 experiments check only down to the 3σ level (Appendix F.4).

Further material is in Appendix F.

VIII. LOCAL ASYMMETRY IN THE DALITZ PLANE

In short: the asymmetry is not uniform over the Dalitz plane ($\chi^2 = 80.8$ for 31 *degrees of freedom); it is concentrated at low m^2_{lo} outside the $\phi(1020)$ and, in the region of the published analysis of the same data, it reproduces the published value.* To test whether the asymmetry is uniform over phase space or concentrated in specific regions, each candidate was placed in the Dalitz plane using the two opposite-sign K^+K^- invariant masses $m^2(H1, H2)$ and $m^2(H1, H3)$ (Sec. II A), sorted into m^2_{lo} and m^2_{hi} to remove the $H2/H3$ labeling ambiguity. The plane $[0, 16] \times [0, 30]$ GeV^2 in $(m^2_{\text{lo}}, m^2_{\text{hi}})$ was divided into an

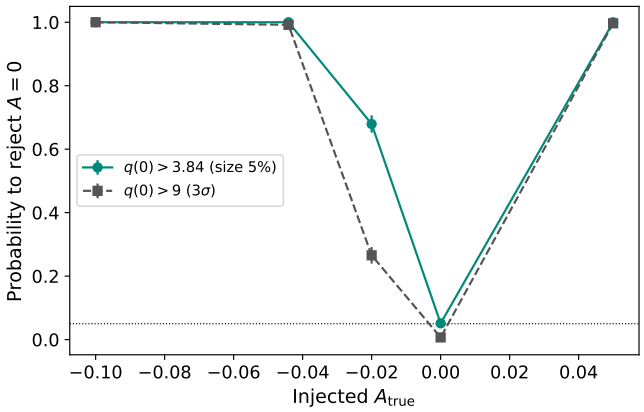


FIG. VI. Probability to reject $A = 0$ as a function of the injected asymmetry, with the test $q(0) > 3.84$ (size 5%, dotted line) and $q(0) > 9$ (3σ); 1000 toy experiments at -0.0441, about 300 at each other value.

8×8 grid, fixed before looking at the asymmetries. In each bin with at least 20 candidates (32 of 64), the local asymmetry $A_i = (N_i^- - N_i^+)/N_i$, with binomial uncertainty $\sigma(A_i) = \sqrt{(1 - A_i^2)/N_i}$, was compared with the global counting asymmetry $A_{\text{glob}} = -0.043$ through $S_i = (A_i - A_{\text{glob}})/\sigma(A_i)$.

Figure VII shows the occupancy of the bins (a), the local asymmetry (b) and its significance (c). Two of the 32 bins have $|S_i| > 3$, adjacent at $m^2_{\text{lo}} < 4$ GeV^2 and $7.5 < m^2_{\text{hi}} < 11.25$ GeV^2 ($S_i = -3.7$ and -3.0); about 0.09 such bins would be expected if the asymmetry were uniform, and two or more with probability 3×10^{-3}, a look-elsewhere number over any two of the 32 bins. The global homogeneity test, $\chi^2 = \sum_i S_i^2 = 80.8$ for 31 degrees of freedom ($p = 3 \times 10^{-6}$), confirms that the local asymmetry is not uniform; it also holds in the signal window alone, with the charm vetoes and for other grids (Appendix G).

The structure is the same with both magnet polarities: in the region of the two significant bins the counting asymmetry is -0.129 ± 0.023 (magnet down) and -0.126 ± 0.028 (magnet up), and the two maps agree bin by bin ($\chi^2 = 23.0$ for 29 bins, $p = 0.78$). The split tests the part of the detection asymmetry that changes sign with the polarity, not the different interaction of K^+ and K^- with the detector material.

Where is the asymmetry in terms of the decay dynamics? The ordered Dalitz plot (Fig. VIIIa) shows the $\phi(1020)$ band at the kinematic edge ($m^2_{\text{lo}} \approx 1.04$ GeV^2) and the charm bands of the D^0, J/ψ and χ_{c0}. The negative asymmetry is not produced by the $\phi(1020)$: its band (1964 candidates) has $A_{\text{count}} = -0.004 \pm 0.023$, while the significant bins without it have -0.145 ± 0.019 (Fig. XXVI of Appendix G). It sits just above the ϕ, where the published analysis located it, defining the region $1.2 < m^2_{\text{lo}} < 2.0$ GeV^2, $m^2_{\text{hi}} < 15$ GeV^2 with $A_{\text{raw}} = -0.239 \pm 0.020$ [5]. The region is fixed by the paper, not chosen on our data, so the comparison is unbiased: the nominal model fitted to its 2146 candidates gives $A_{\text{raw}} = -0.234 \pm 0.025$ (Bayesian) and -0.237 ± 0.025 (frequentist, 9.4σ from zero), reproducing the published value (spectra of the two charges in Fig. VIIIb, where the B^- peak is visibly smaller).

Three caveats limit the interpretation: the binning is coarse and was not optimized, the asymmetries are counts without sWeight background subtraction (Appendix J.5), and local variations can be partly instrumental, since the $K^\pm$ interaction asymmetry depends on momenta that are correlated with the Dalitz position. In summary, the global asymmetry is not uniform over phase space, and the region identified by the published analysis carries it.

Further material is in Appendix G.

IX. COMPARISON AND DISCUSSION

In short: the frequentist and Bayesian analyses give the same value, uncertainty and coverage for A_{raw} because they use the same likelihood and the data constrain it well; they differ in what the numbers mean and in the hypothesis tests, and a machine-learning selection (Appendix I) confirms the result without improving it.

The frequentist and Bayesian analyses agree on every number and differ in what each number means. Table IV puts the two approaches side by side, from the question each one asks to how it was validated; the subsections explain the rows where the numbers coincide but the statements do not.

A. Where the Approaches Differ

In short: the numbers coincide but the statements differ: the meaning of the interval, the test of $A = 0$, the goodness of fit, the treatment of the nuisance parameters and what has been validated, as summarized in Table IV. Meaning of the interval. The credible interval states that, given the data and the prior, A lies in $[-0.0528, -0.0358]$ with 68% probability; the confidence interval $[-0.0526, -0.0356]$ is the set of values that the data do not reject at the 32% level, and the 68% is the fraction of repeated experiments in which such an interval contains the true value, a property of the procedure that the bootstrap and toy checks verified. Figure IX shows the two for A: the profile likelihood and the marginal posterior coincide (a), and the profile curves of the two magnet polarities add up to that of the joint fit (b).

Hypothesis test. Three tests of $A = 0$ are available. The exact binomial test of $N^- = N^+$ on all candidates gives 6.1σ ($A_{\text{count}} = -0.0432 \pm 0.0071$); it sees signal and background together and does not use the mass. The frequentist test of the fit gives $p = 2 \times 10^{-7}$ ($Z = \sqrt{q(0)} = 5.2\sigma$ in the two-sided convention, 5.06σ

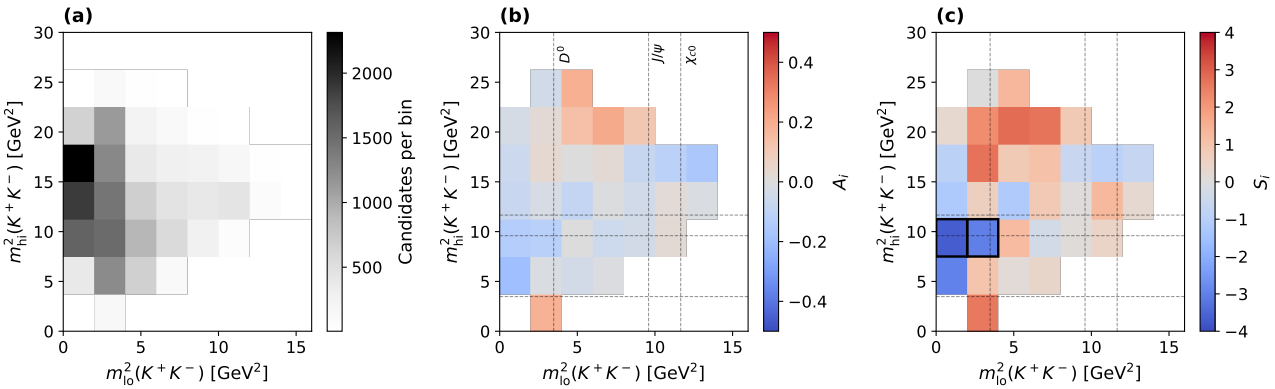


FIG. VII. Fixed 8×8 Dalitz-plot binning, all candidates in the fit window: (a) number of candidates per bin, (b) local asymmetry A_i for bins with ≥ 20 candidates, and (c) local significance with respect to the global asymmetry, $(A_i - A_{\rm glob})/\sigma(A_i)$. The two bins with $|S_i| > 3$ are outlined. Dashed lines: the squared masses of the D^0, J/ψ and χ_{c0}, in either opposite-sign pair.

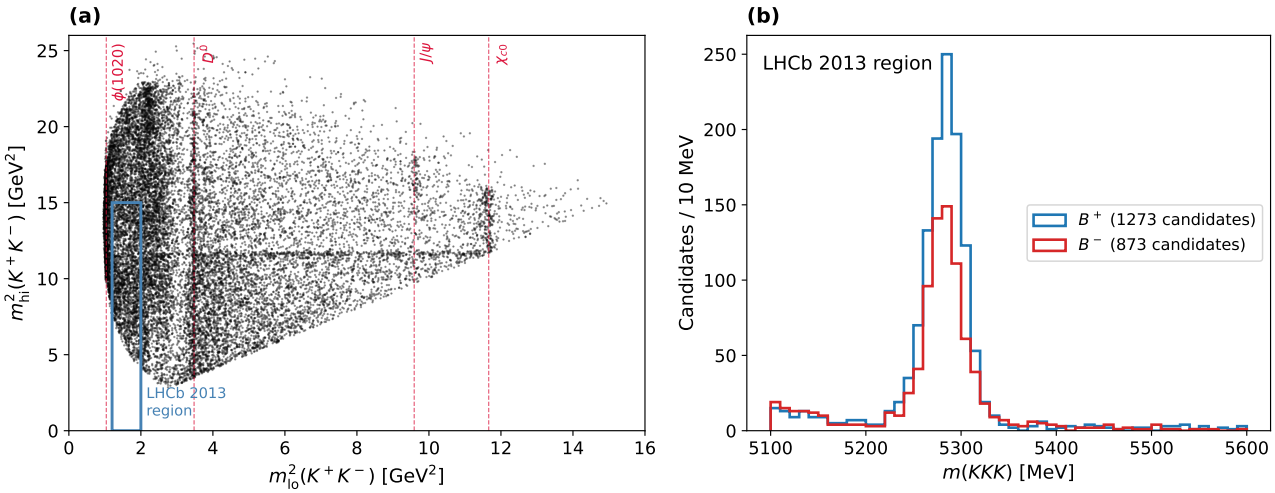


FIG. VIII. (a) Ordered Dalitz plot of the candidates in the fit window, with the squared masses of the $\phi(1020)$, D^0, J/ψ and χ_{c0} (dashed) and the region of the published analysis of the same data [5] (box). (b) Mass spectra of B^+ and B^- candidates in that region, same binning: the B^- peak is visibly smaller.

one-sided, Appendix C.1), independent of the prior and not the probability of $A = 0$; its size and power were verified with toy MC (Appendix F.4).

The Bayes factor, $BF_{10} \approx 9\times10^3$, depends on the prior width (Sec. V D). For A the frequentist and Bayesian tests give the same conclusion; for the background asymmetry, $A_b = -0.040 \pm 0.018$, a frequentist test would call 2.3σ a hint while $BF_{01} \approx 3.5$ mildly favors $A_b = 0$: the p-value asks how improbable the data are under $A_b = 0$, the Bayes factor which hypothesis the data prefer after the prior volume is accounted for.

B. Comparison with the Published Analysis

In short: the published value (-0.060 ± 0.007) *differs from this one* (-0.0441 ± 0.0085) *by the sample and the model; with the same* D^0 *veto the Bayesian fit gives* -0.0496 ± 0.0088*, and* $A_{\rm raw}$ *is not* A_{CP}*.* The published analysis [5] finds $A_{\rm raw} = -0.060 \pm 0.007$ with $22{,}119 \pm 164$ signal candidates, against -0.0441 ± 0.0085 and $15{,}328 \pm 133$ here: the Open Data record is a preselected subset and the published model has a Cruijff signal, partially reconstructed four-body decays and a misidentified peaking background from simulation. With the same D^0 veto the Bayesian fit gives -0.0496 ± 0.0088, within 0.93σ of the published value.

The raw asymmetry is not the CP asymmetry: $A_{CP} \approx A_{\rm raw} - A_P - A_D$ (Eq. (2)), and in the published analysis the corrections move the result from -0.060 to $-0.043 \pm 0.009 \pm 0.003 \pm 0.007$ (3.7σ) [5], by $+0.017$, about two statistical uncertainties of this work. Such a correction cannot be assumed for a different selection, and the 5.2σ here is the significance of a nonzero raw asymmetry, not of CP violation. A later measurement on a larger data set gives $A_{CP} = -0.037 \pm 0.002 \pm 0.002 \pm 0.003$ (8.5σ) [1, 2];

	Frequentist (Sec. IV)	Bayesian (Sec. V)
Question	Which value of A makes the data most probable, and which values are not rejected?	What is the distribution of A given the data and the prior?
Estimate	Maximum likelihood, -0.0441	Posterior median, -0.0443 (mode -0.0441)
Interval (68%)	$[-0.0526, -0.0356]$ (profile likelihood, Minos)	$[-0.0528, -0.0358]$ (credible, equal-tailed)
Nuisance parameters (11)	profiled: re-minimized at each fixed A	marginalized: integrated over the prior
Test of $A = 0$	$q(0) \simeq 27$, $p = 2.05 \times 10^{-7}$, 5.2σ	$BF_{01} \approx 1.1 \times 10^{-4}$ (Savage-Dickey)
Validation of the interval	1000 bootstrap replicas: $68.4 \pm 1.5\%$; 985 toys: $70 \pm 1\%$ (68%), $95 \pm 1\%$ (95%)	200 toys: 0.65 ± 0.03 (68%), 0.93 ± 0.02 (95%)
Goodness of fit	χ^2/ndof $\approx$ 160/88 (10 MeV bins); 441.7/388, $p = 0.031$ (2.5 MeV bins)	164.1/88 ($p = 1.6 \times 10^{-6}$); posterior predictive check
Computing time	one joint fit with Minos $\simeq 19$ s; 1000 bootstrap fits $\simeq 49$ min	one chain (25,000 steps, 32 walkers) $\simeq 8$ min; 200 toys of 12,000 steps

TABLE IV. Frequentist and Bayesian analyses compared on the same candidates (A_{raw}, statistical uncertainties only). Computing times are indicative, on one laptop.

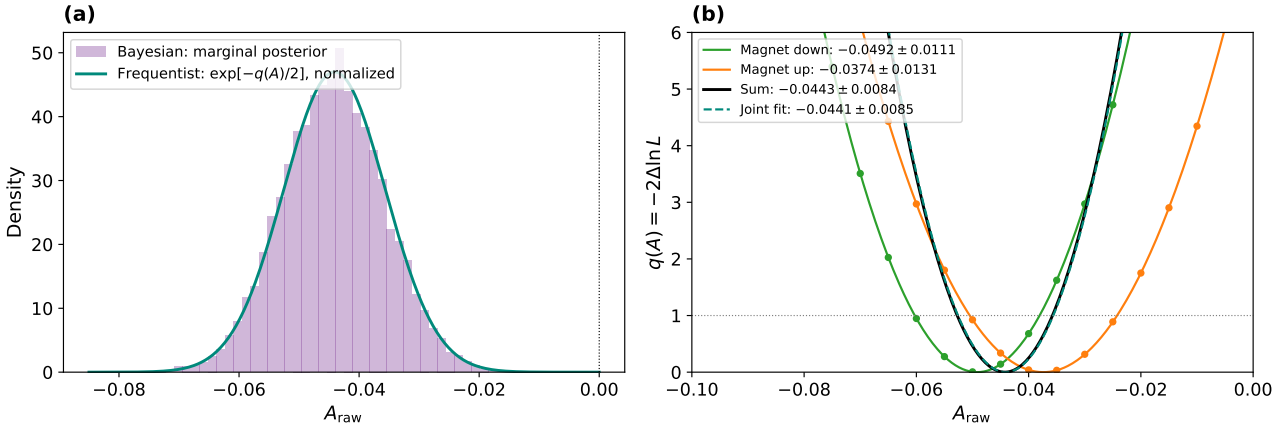


FIG. IX. (a) Frequentist profile likelihood, $\exp[-q(A)/2]$ normalized, and Bayesian marginal posterior of A. (b) Profile curves $q(A)$ of the two magnet polarities (parabolic fits to the points), their sum and the joint fit.

our result is consistent with these in sign and size.

C. Machine-Learning Cross-Check

In short: a classifier trained on weak labels from the mass spectrum selects candidates without changing A_{raw} beyond its statistical spread; it does not improve the precision and cannot remove the misidentified decays under the signal (Appendix I). A cut at 0.5 on the classifier score keeps 92% of the fitted signal and 36% of the background and gives $A_{\text{raw}} = -0.0451 \pm 0.0087$ (Bayesian) and -0.0447 ± 0.0086 (frequentist), against -0.0443 and -0.0441 for the full window: both shifts are within the statistical spread expected for a subsample (Appendix I.9, Fig. XXXIX). The precision does not improve because the nominal sample has little background to remove; the classifier is described in Appendix I.

Further material is in Appendix H.

X. CONCLUSIONS

In short: two inference frameworks give the same raw asymmetry, $A_{\text{raw}} = -0.044 \pm 0.009\,(stat) \pm 0.007\,(syst)$, 5.2 standard deviations from zero, with validated intervals; what separates them is the meaning of the numbers and the tests, not their values; a machine-learning selection (Appendix I) confirms the result without improving it.

The answer to the question of the Introduction is that, on this problem, the frequentist profile-likelihood fit and the Bayesian posterior are equivalent in value, uncertainty and coverage. With a common selection and mass model on the 2011 LHCb Open Data of $B^{\pm} \to K^{\pm}K^{+}K^{-}$ they give $A_{\text{raw}} = -0.0441 \pm 0.0085 \pm 0.0072$ and $-0.0443 \pm 0.0085 \pm 0.0070$ (statistical, systematic), a nonzero asymmetry at 5.2 standard deviations, stable under the thirteen variants of the fit and selection, compatible in sign and size with the published analysis, and concentrated in its Dalitz region outside the $\phi(1020)$. It is a raw asymmetry, not a measurement of A_{CP} and not

a significance of CP violation (Sec. IX B).

The methods differ in what they add: the frequentist fit an unbiased estimator at the minimum variance bound with intervals of nominal coverage; the Bayesian analysis the same interval with flat priors, plus sampling diagnostics, a Bayes factor that depends on the prior width and a posterior predictive check. The machine-learning selection leaves $A_{\rm raw}$ unchanged. The dominant systematic contribution is the particle-identification requirement, in both methods. Three limitations bound the result (Appendix J.5): no tested model reaches $\chi^2/\mathrm{ndof} \approx 1$, although $A_{\rm raw}$ does not depend on the signal shape; no production or detection correction is applied, so $A_{\rm raw} \neq A_{CP}$; and the local Dalitz-plane asymmetries are counts without background subtraction by event weights.

Appendix A: Dataset and Preselection

In short: the record and its branches, the cut flow of the preselection, and the candidates and charge combinations that the main text treats in one line.

1. Description of the Record

In short: the record holds 8,556,118 *candidates of* $B^\pm \to h^\pm h^+ h^-$ *collected in 2011, in two files by magnet polarity, plus a detector-free phase-space simulation of* $B^\pm \to K^\pm K^+ K^-$ *used only to test the local-asymmetry procedure.*

2. Branches and Preselection

In short: the analysis reads momentum, particle-identification likelihoods, muon flag and charge of each track; the common preselection keeps 23,895 *candidates (*0.28%*), mostly through the kaon-identification requirement.*

3. Cut Flow of the Preselection

In short: the kaon-identification requirement on the three tracks is by far the most selective.

Requirement	alone	cumulative	down	up
All candidates	–	8,556,118	5,135,823	3,420,295
¬`isMuon`, three tracks	6,311,517	6,311,517	3,806,297	2,505,220
`ProbPi` < 0.5, three tracks	975,724	481,101	288,908	192,193
`ProbK` > 0.5, three tracks	28,812	23,895	14,178	9717
Mass window $[5100, 5600]$ MeV	–	19,731	11,636	8095

TABLE V. Number of candidates passing each requirement alone and after the cumulative selection, in the order of the table, and by magnet polarity. The kaon-identification requirement on the three tracks is by far the most selective.

4. Candidates Sharing a Track

In short: only 42 *candidates of the fit window (*0.2%*) share a track with another candidate, a negligible effect, and all are kept.*

5. Charge Combinations

In short: $H1$ *is always the kaon of charge opposite to the* B *and* $H2$, $H3$ *the same-sign pair, so the Dalitz variables are the two opposite-sign invariant masses.*

6. Invariant Mass and Fit Window

In short: the mass is computed with the kaon hypothesis on all tracks; the fit window $[5100, 5600]$ *MeV holds* 19,731 *candidates and includes the start of the rise of the background below* ~ 5150 *MeV.*

Appendix B: Mass Model and Likelihood

In short: the extended likelihood and its relation to a mixture, the label switching of the two background slopes, and the six-parameter model that preceded the nominal one.

1. Extended Likelihood

In short: with a Poisson total number of candidates the likelihood is $e^{-(N_s+N_b)} \prod_i [N_s S + N_b B]$ *up to a constant, which constrains the yields and* A *together.*

2. Extended Likelihood versus a Mixture with Fixed Total

In short: the extended form differs from a mixture with fixed total only by the Poisson term of the total yield, which carries no information on A*; the purity of the signal window is about* 96%*.*

3. Label Switching of the Background Components

In short: the mixture is invariant under the exchange of the two exponentials, which would give two identical modes; the ordering $\lambda_1 > \lambda_2$ *removes the symmetry.*

4. Minimal Model as a Cross-Check

In short: a six-parameter model on a narrower window gives $A_{\mathrm{raw}} = -0.0497$ *(68% CI* $[-0.058, -0.041]$*), compatible with the nominal result within its uncertainty; it is the model used for the low-statistics comparison.*

Appendix C: Frequentist Analysis

In short: what the Hesse and Minos uncertainties and the test of $A = 0$ *are, the Neyman construction, the uncertainties of the twelve parameters by three methods, and the behavior of the fit from different starting values.*

1. Hesse Uncertainties, Test of $A = 0$ and Significance

In short: the Hesse matrix is the inverse of the Fisher information, the test statistic is the profile likelihood ratio, and the significance is $Z = \sqrt{q(0)}$*.*

2. Neyman Construction and Dependence on the Momentum

In short: inverting the test of A_{true} *with toy-MC experiments gives the same interval as the likelihood ratio, and* A_{raw} *does not depend on the momentum of the* B*.*

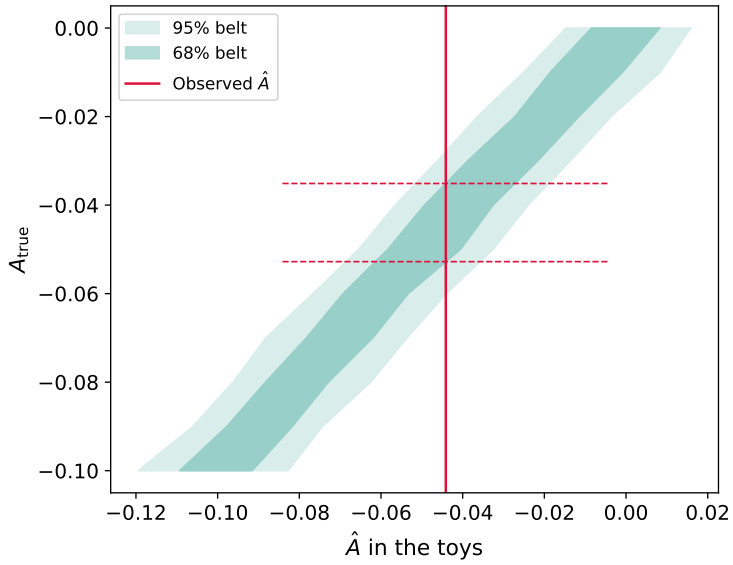


FIG. X. Confidence belt of $\hat{A}$ from toy-MC experiments (central 68% and 95% ranges) as a function of A_{true}; the vertical line is the observed $\hat{A}$ and the dashed lines the resulting 68% interval.

Bin	range [GeV]	n	A_{raw}
p_T, 1	1.0–4.7	6577	-0.0563 ± 0.0169
p_T, 2	4.7–8.2	6577	-0.0362 ± 0.0141
p_T, 3	8.2–30.5	6577	-0.0452 ± 0.0134
p, 1	26.7–78.1	6577	-0.0391 ± 0.0139
p, 2	78.1–110.1	6577	-0.0475 ± 0.0149
p, 3	110.1–288.5	6577	-0.0478 ± 0.0153

TABLE VI. A_{raw} (joint frequentist fit) in terciles of the transverse momentum and of the momentum of the B candidate.

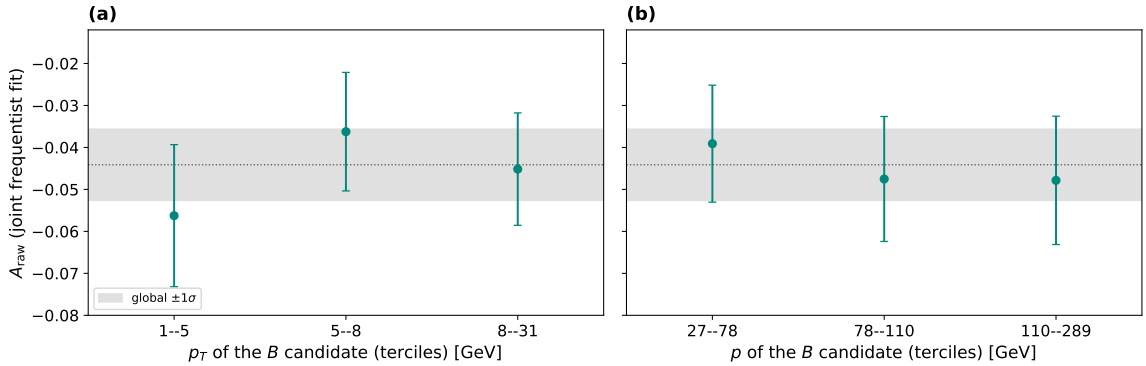


FIG. XI. A_{raw} (joint frequentist fit) in terciles of (a) the transverse momentum and (b) the momentum of the B candidate, with the global value and its $\pm 1\sigma$ band; each bin has 6577 candidates.

3. Uncertainties of the Twelve Parameters by Three Methods

In short: the inverse of the Fisher information (Hesse), the bootstrap of the data and the posterior give the same standard deviations for A, A_b, μ and σ; the bootstrap differs for the two yields, because it keeps the total number of candidates fixed, and for the steep slope λ_1^+, which is weakly constrained.

Parameter	Hesse	Bootstrap	Posterior
A	0.0085	0.0085	0.0085
A_b	0.0177	0.0183	0.0176
N_s	133	72	133
N_b	82	72	82
μ [MeV]	0.159	0.159	0.161
σ [MeV]	0.141	0.150	0.138
λ_1^+	0.0077	0.0057	0.0078
λ_2^+	0.00028	0.00030	0.00029
f_+	0.028	0.029	0.030
λ_1^-	0.0047	0.0043	0.0047
λ_2^-	0.00040	0.00038	0.00039
f_-	0.039	0.036	0.037

TABLE VII. Standard deviation of the parameters of the nominal fit from the Hesse matrix, from the bootstrap of the data and from the posterior.

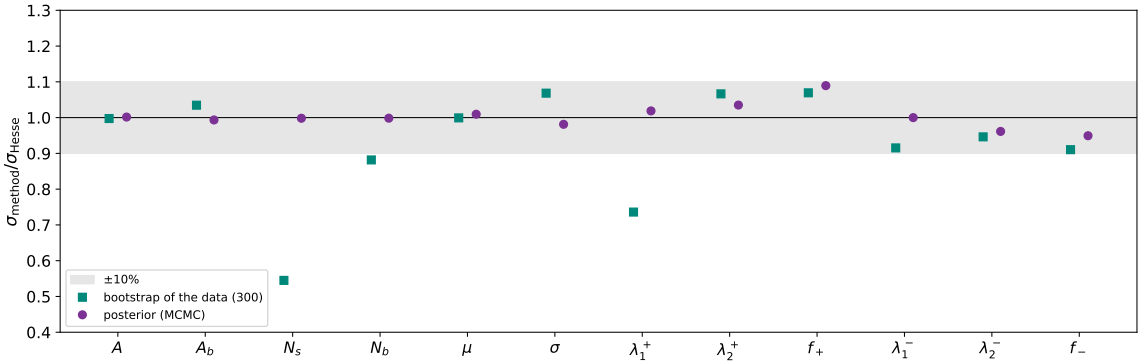


FIG. XII. Standard deviation of the twelve parameters of the nominal fit from the bootstrap of the data (300 replicas) and from the posterior, divided by the Hesse uncertainty; the band is ±10%.

4. Starting Values and Weakly Constrained Parameters

In short: 29 *of* 30 *fits from random starting points reach the same minimum; the steep-background fraction is the weakly constrained direction, as the long chain of the Bayesian fit already showed.*

Appendix D: Bayesian Analysis

In short: the sampling and the posterior summaries behind Sec. V, the Bayes factors with their derivation, the model-adequacy and robustness checks, the toy-MC coverage of the credible interval and the alternative sampler.

1. Sampling the Posterior

In short: Metropolis-Hastings accepts a proposal with a ratio that satisfies detailed balance; the affine-invariant ensemble move samples correlated posteriors without tuning, and N/τ is the effective length of a chain.

2. Posterior Summaries and Model Comparison

In short: equal-tailed and HPD intervals, the Jeffreys scale for Bayes factors, the Ockham effect of the prior width and the definition of the posterior predictive p-value.

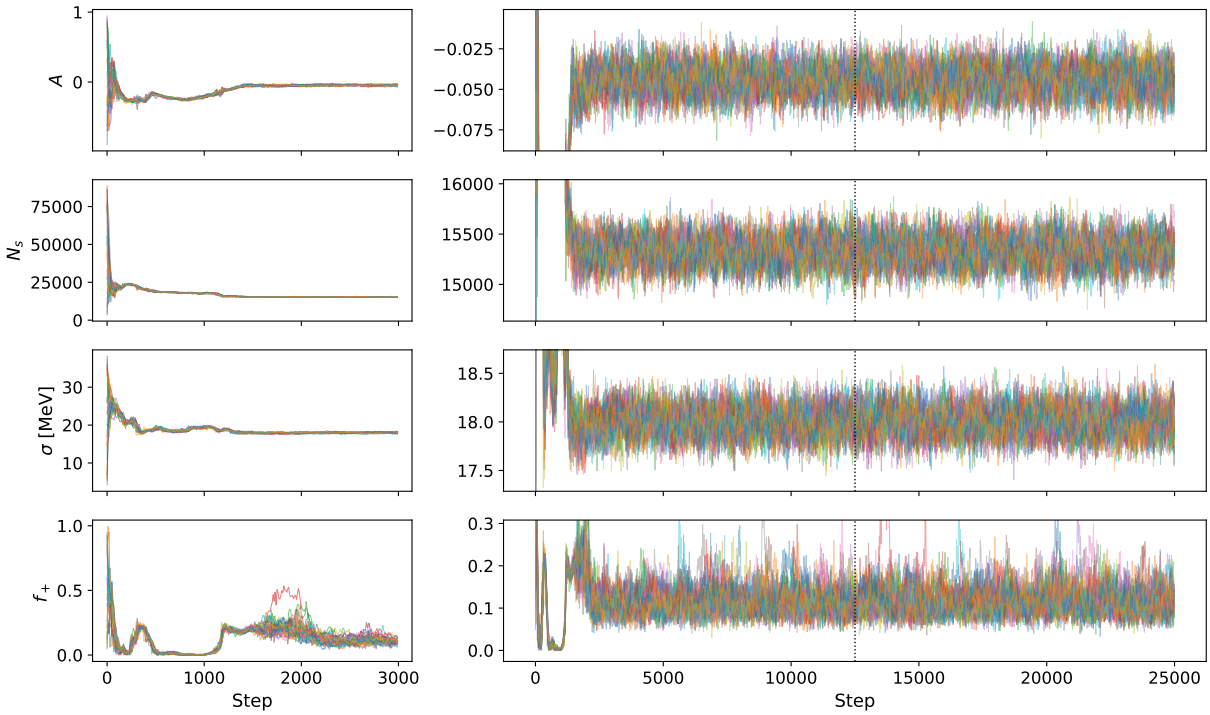


FIG. XIII. Trace plot of the nominal fit with the 32 walkers initialized uniformly over the prior box (one line per walker, every 10th step), for A, N_s, σ and the steep-background fraction f_+. Left: first 3000 steps, full range; right: whole chain, zoomed on the posterior. The first half of the chain (dotted line) is discarded as burn-in.

3. Flat Priors and the Evidence

In short: a flat prior is not invariant under reparametrization, but with $\sim 2 \times 10^4$ *events the posterior is dominated by the likelihood; the evidence is a constant that neither the MCMC nor the Savage-Dickey ratio needs.*

4. Sampling and Convergence Diagnostics

In short: 32 *walkers for* 25,000 *steps give* $N/\tau \geq 113$ *autocorrelation times for every parameter, and runs started over the whole prior box converge to the same single mode.*

5. Trace Plots of the Walkers

In short: walkers started over the whole prior box converge to the same mode.

6. Posterior of the Raw Asymmetry

In short: $A_{\mathrm{raw}} = -0.0443$ *with* 68% *and* 95% *intervals* $[-0.0528, -0.0358]$ *and* $[-0.0608, -0.0276]$*; the posterior is Gaussian, so equal-tailed and highest-density intervals coincide and its mode is the maximum-likelihood estimate.*

7. Equal-Tailed and Highest-Posterior-Density Intervals

In short: for A *and* A_b *the two kinds of* 68% *interval coincide; they differ for the skewed parameters, whose highest-posterior-density interval is shifted towards the mode.*

Parameter	median	equal-tailed	HPD	skewness
A	-0.0442	$[-0.0529, -0.0358]$	$[-0.0525, -0.0355]$	$+0.01$
A_b	-0.0400	$[-0.0575, -0.0224]$	$[-0.0575, -0.0225]$	$+0.01$
f_+	0.114	$[0.089, 0.145]$	$[0.084, 0.138]$	$+0.90$
f_-	0.209	$[0.174, 0.248]$	$[0.168, 0.241]$	$+0.33$
λ_1^+ [MeV^{-1}]	0.0350	$[0.0281, 0.0431]$	$[0.0274, 0.0423]$	$+0.52$

TABLE VIII. Equal-tailed and highest-posterior-density 68% intervals and skewness of the marginal posteriors.

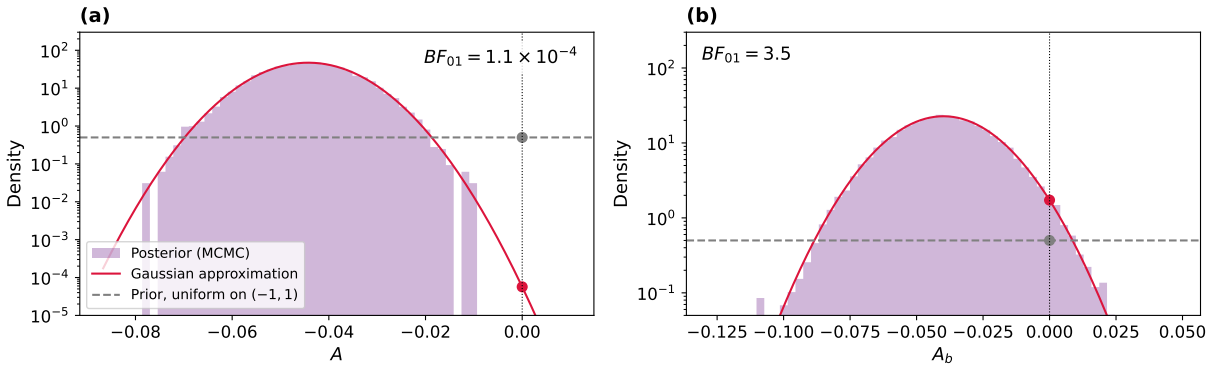


FIG. XIV. Savage-Dickey ratio for (a) A and (b) A_b: marginal posterior (histogram of the chain and Gaussian approximation) and prior density, logarithmic scale. The Bayes factor BF_{01} is the ratio of the two densities at zero (dots). For A the posterior density at zero is 10^{-4} times the prior one; for A_b, 2.3 standard deviations from zero, it is higher than the prior density, and the data favor $A_b = 0$.

8. Savage–Dickey Ratio

In short: the Bayes factor of $A = 0$ is the ratio of the posterior and prior densities at $A = 0$, with no integral over the twelve parameters.

9. Bayes Factor of $A = 0$

In short: the Savage-Dickey ratio with the Gaussian approximation of the posterior gives $BF_{01} \approx 1.1 \times 10^{-4}$ ($BF_{10} \approx 9 \times 10^3$), decisive evidence against $A = 0$ that scales with the prior width.

10. Bayes Factor of the Background Asymmetry

In short: $A_b = -0.040 \pm 0.018$ is 2.3 standard deviations from zero, yet $BF_{01} \approx 3.5$ favors $A_b = 0$ (Ockham factor); A_b is kept free in the nominal model.

11. Savage–Dickey Densities of the Parameters

In short: the posterior and prior densities at the null value, which give the Bayes factors of the main text.

12. Priors and Bayes Factors

In short: no change of the prior on the nuisance parameters or of its parametrization moves A by more than 0.0002; the Bayes factor of $A = 0$ scales with the prior width, and the evidence computed by direct integration agrees in order of magnitude with the Savage-Dickey ratio.

Prior	A (median)	68% interval	ESS
Nominal (uniform)	-0.0442	$[-0.0529, -0.0358]$	20000
Log-uniform N_s, N_b	-0.0442	$[-0.0529, -0.0358]$	19993
Log-uniform slopes	-0.0440	$[-0.0528, -0.0356]$	12306
Uniform in $1/\lambda$	-0.0444	$[-0.0526, -0.0360]$	153
Flat in (N_s^+, N_s^-)	-0.0442	$[-0.0529, -0.0358]$	19999
Flat in $(N_s^\pm)$ and $(N_b^\pm)$	-0.0442	$[-0.0529, -0.0358]$	19993

TABLE IX. Posterior of A under alternative priors, by importance reweighting of the chain; ESS: effective sample size of the 20,000 thinned samples.

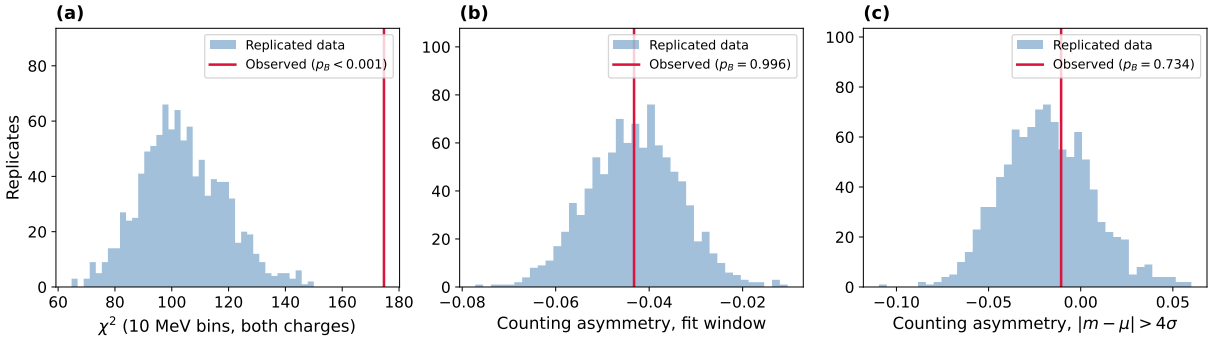


FIG. XV. Posterior predictive check: distributions of three discrepancies over 1000 datasets replicated from posterior draws, compared with the observed values. (a) binned χ^2; (b) and (c) counting asymmetry of the fit window and of the sidebands.

13. Model Adequacy: Goodness of Fit and Posterior Predictive Check

In short: the model does not reach $\chi^2/\text{ndof} \approx 1$ *(*$164.1/88$*), mainly because the signal tails are heavier than a Gaussian, but* A_{raw} *changes by only* -0.0002 *with a Crystal Ball signal.*

Model	A_{raw}	χ^2/ndof	p
Nominal	-0.0443	$164.1/88$	2×10^{-6}
Crystal Ball signal	-0.0445	$131.0/86$	1×10^{-3}
μ, σ per charge	-0.0443	$164.2/86$	8×10^{-7}
Shared bkg. shape	-0.0480	$174.1/91$	4×10^{-7}
Symmetric bkg. ($A_b = 0$)	-0.0471	$169.8/89$	5×10^{-7}
Single exponential	-0.0480	$297.7/92$	2×10^{-23}

TABLE X. A_{raw} (posterior median) and goodness of fit of the nominal model (Gaussian signal, double exponential per charge) and of the alternative signal and background models, same data and window.

14. Posterior Predictive Check with Statistics Not Fitted by the Model

In short: the asymmetry in slices of the mass, which the model predicts, is reproduced ($p_B = 0.67$*), and the difference between the* χ^2 *of the two charges is at the edge of the replicas (*$p_B = 0.08$*).*

15. Robustness Checks: Polarity, Prior and Charm Vetoes

In short: A_{raw} *agrees between the two magnet polarities (*0.7σ*) and does not depend on the prior; the* D^0 *veto moves it by* 2.3 *times the expected statistical spread.*

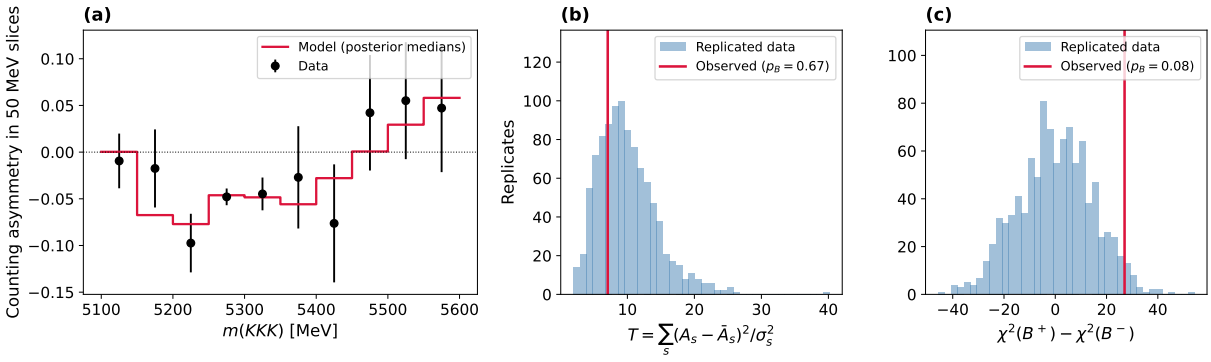


FIG. XVI. (a) Counting asymmetry in slices of 50 MeV of the mass, data and model. (b) Distribution of $T = \sum_s (A_s - \bar{A}_s)^2/\sigma_s^2$ over 1000 replicas and observed value. (c) The same for $\chi^2(B^+) - \chi^2(B^-)$.

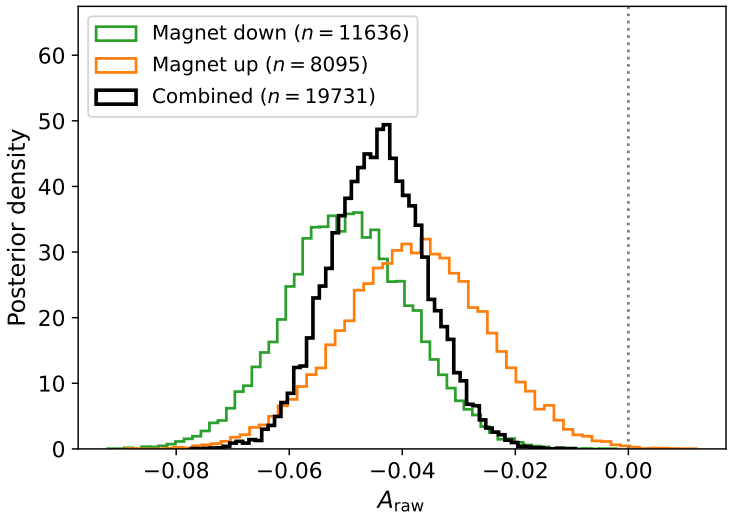


FIG. XVII. Posterior of A_{raw} for the magnet-down, magnet-up and combined samples.

16. Toy-MC Coverage Validation

In short: 200 *toy-MC experiments give coverage* 0.65 ± 0.03 *(68%) and* 0.93 ± 0.02 *(95%), consistent with nominal, with unbiased estimates and pulls of width* 1.04.

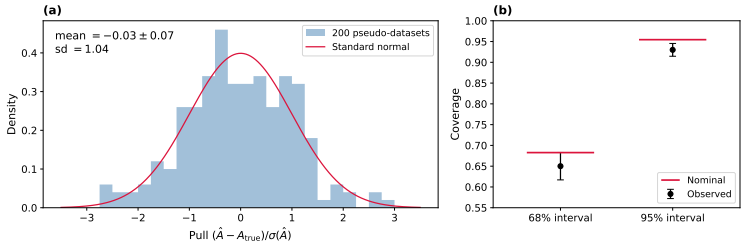


FIG. XVIII. Toy-MC validation, 200 toy-MC datasets. (a) Pulls $(\hat{A} - A_{\text{true}})/\sigma(\hat{A})$, with $\hat{A}$ and $\sigma(\hat{A})$ the posterior median and standard deviation of each replica, compared with a standard normal. (b) Fraction of replicas whose 68% and 95% credible intervals contain the injected value, with binomial uncertainties, and the nominal coverage.

17. Metropolis-Hastings Sampler and Published Likelihood

In short: an own Metropolis-Hastings sampler reproduces the posterior of the ensemble sampler, and the profile likelihood is given in a table, so that the result can be combined with others.

A	-0.08	-0.07	-0.06	-0.05	-0.04	-0.03	-0.02	-0.01	$+0.00$
$q(A)$	17.9	9.3	3.5	0.5	0.2	2.8	8.1	16.1	27.0

TABLE XI. Profile likelihood ratio $q(A) = -2\Delta \ln L$ of the nominal fit (other parameters profiled); minimum at -0.0441, Minos intervals $[-0.0526, -0.0356]$ (68%) and $[-0.0607, -0.0275]$ (95%).

Appendix E: Systematic Uncertainties

In short: every fit of the nominal model and of its variants with both methods, the variants and the rule that combines them, and the validation of that rule on nested samples.

1. All Fits of the Nominal Model and of Its Variants

In short: the Bayesian and frequentist results of every variant and cross-check side by side.

Fit	n	A_{raw} Bayesian (median)	68% CI	χ^2/ndof	A_{raw} frequentist
Nominal	19731	-0.0443	$[-0.0528, -0.0358]$	164.1/88	-0.0441 ± 0.0085
Systematic variants (Sec. VI)					
Window $[5050, 5600]$	21164	-0.0446	$[-0.0531, -0.0362]$	217.4/98	-0.0445 ± 0.0085
Window $[5150, 5600]$	18566	-0.0444	$[-0.0528, -0.0359]$	141.5/78	$-0.0446 \pm 0.0088^{\dagger}$
Window $[5100, 5450]$	19004	-0.0440	$[-0.0524, -0.0357]$	136.9/58	-0.0442 ± 0.0085
Window $[5100, 5700]$	20191	-0.0456	$[-0.0540, -0.0373]$	200.2/108	-0.0455 ± 0.0085
Single exponential	19731	-0.0480	$[-0.0564, -0.0395]$	297.7/92	-0.0481 ± 0.0085
Background shape shared	19731	-0.0480	$[-0.0564, -0.0397]$	174.1/91	-0.0481 ± 0.0083
Symmetric background	19731	-0.0471	$[-0.0552, -0.0388]$	169.8/89	-0.0472 ± 0.0084
Crystal Ball signal	19731	-0.0445	$[-0.0529, -0.0363]$	131.0/86	-0.0447 ± 0.0085
μ, σ per charge	19731	-0.0443	$[-0.0527, -0.0356]$	164.2/86	-0.0442 ± 0.0087
`ProbK` > 0.4	26553	-0.0474	$[-0.0550, -0.0399]$	175.6/88	-0.0473 ± 0.0076
`ProbK` > 0.6	13981	-0.0385	$[-0.0482, -0.0289]$	145.0/88	-0.0384 ± 0.0099
`ProbPi` < 0.4	17984	-0.0421	$[-0.0506, -0.0333]$	150.1/88	-0.0421 ± 0.0088
`ProbPi` < 0.6	20953	-0.0446	$[-0.0529, -0.0366]$	159.0/88	-0.0448 ± 0.0083
Cross-checks (Secs. V B, D.15)					
Dispersed start, run 1	19731	-0.0446	$[-0.0529, -0.0360]$	164.1/88	–
Dispersed start, run 2	19731	-0.0442	$[-0.0528, -0.0356]$	164.1/88	–
Magnet down	11636	-0.0497	$[-0.0607, -0.0386]$	153.1/88	–
Magnet up	8095	-0.0378	$[-0.0507, -0.0251]$	150.3/88	–
Narrow prior, $A \in (-0.3, 0.3)$	19731	-0.0443	$[-0.0528, -0.0358]$	164.1/88	–
D^0 veto	18744	-0.0496	$[-0.0584, -0.0409]$	159.1/88	-0.0493 ± 0.0087
D^0, J/ψ, χ_{c0} vetoes	16367	-0.0423	$[-0.0516, -0.0329]$	140.1/88	-0.0421 ± 0.0094

TABLE XII. All fits of the nominal model and of its variants: number of candidates, Bayesian posterior median and 68% credible interval of A_{raw}, binned goodness of fit (10 MeV bins), and frequentist estimate with its Hesse uncertainty (Sec. IV; fitted on the same candidates; the cross-checks of the Bayesian fits that have no frequentist counterpart in this work are marked –). † Uncertainty scaled from the number of events, since the Hesse matrix of that fit is not accurate. Every fit uses 32 walkers and 25,000 steps.

2. List of the Variants

In short: thirteen variants of the nominal choices, in four groups (window, background, signal, preselection), fitted by both methods on exactly the same candidates.

3. Combination Rule

In short: a variant that changes the sample counts only if its shift exceeds σ_B; variants rejected by the data are excluded, the largest shift of each group is its uncertainty, and the groups are added in quadrature.

4. Results and Particle-Identification Scan

In short: the total systematic uncertainties are 0.0070 *(Bayesian) and* 0.0072 *(frequentist), dominated by the preselection through the single variant* `ProbK` > 0.6 *(*$+0.0058$*,* $1.2\,\sigma_B$*).*

5. Shifts of All Variants

In short: the shifts of the Bayesian and frequentist fits coincide variant by variant; the preselection group dominates, and the single exponential, rejected by the data, is drawn open.

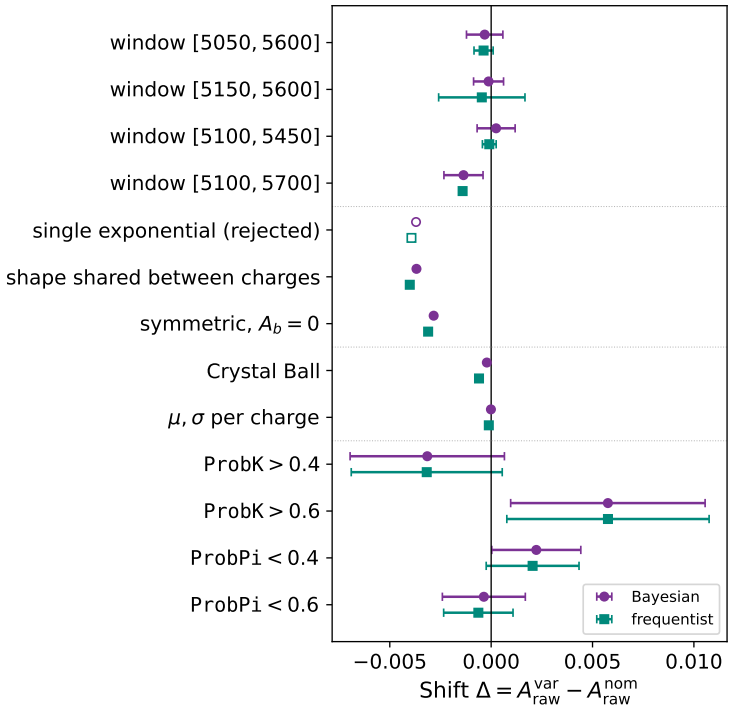


FIG. XIX. Shift $\Delta = A_{\mathrm{raw}}^{\mathrm{var}} - A_{\mathrm{raw}}^{\mathrm{nom}}$ of every variant of the systematic study, Bayesian and frequentist fits; bars: expected spread σ_B of a nested subsample (variants that change the sample only). Open symbols: variant rejected by the data. The dotted lines separate the groups (window, background, signal, preselection).

6. Scan of the Particle-Identification Cuts

In short: on the tight side the shift scatters around zero within $1.3\,\sigma_B$ *and the pion requirement is flat; loosening* `ProbK` *moves* A_{raw} *monotonically, which supports the nominal requirement.*

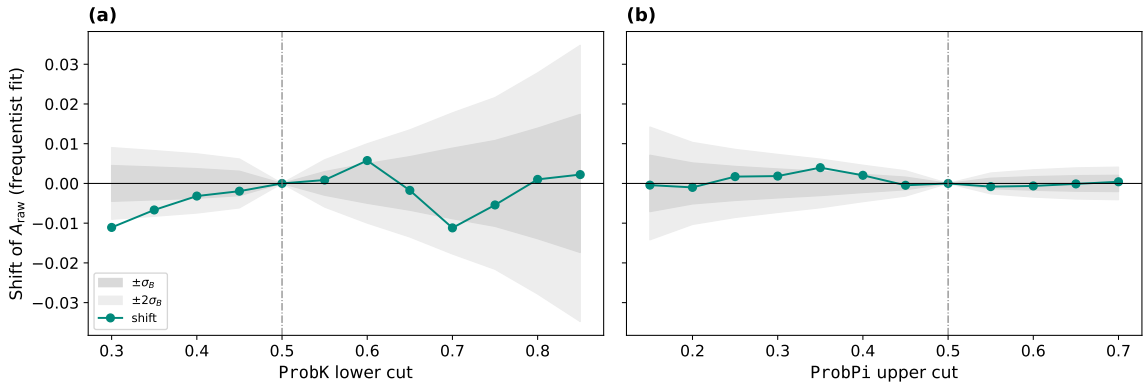


FIG. XX. Shift of A_{raw} (frequentist fit) as a function of (a) the `ProbK` lower cut and (b) the `ProbPi` upper cut, in steps of 0.05, with the expected statistical spread of the nested sample ($\pm\sigma_B$ dark, $\pm2\sigma_B$ light; zero at the nominal cut, dash-dotted line). On the tight side of `ProbK` the points scatter around zero without trend and the pion requirement is flat; loosening `ProbK` moves A_{raw} monotonically.

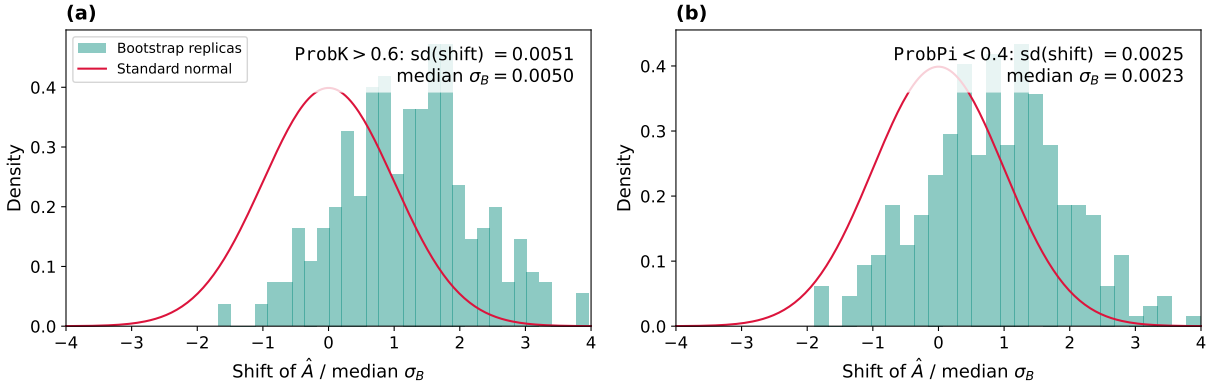


FIG. XXI. Shift of $\hat{A}$ between the nominal sample and a tighter selection, divided by the median σ_B, in 300 bootstrap replicas of the data: (a) `ProbK` > 0.6, (b) `ProbPi` < 0.4, and standard normal distribution.

7. Sensitivity of σ_B to Differences between the Methods

In short: σ_B is the difference of two nearly equal uncertainties and is sensitive to small differences between the methods; a variant at the threshold can be classified differently without changing the result.

8. Validation of the Rule for Nested Samples

In short: the spread of the shift of $\hat{A}$ between a sample and its tighter-identification subsample is the one given by the quadrature difference of the uncertainties.

9. Stability Scans of the Frequentist Fit: Numbers

In short: the largest shifts are 0.0048 *over the grid of window edges and* 0.0126 *over the grid of the two PID cuts, whose samples are smaller (mean uncertainty* 0.0112*); the signal region moves A_{raw} by 6×10^{-5}.*

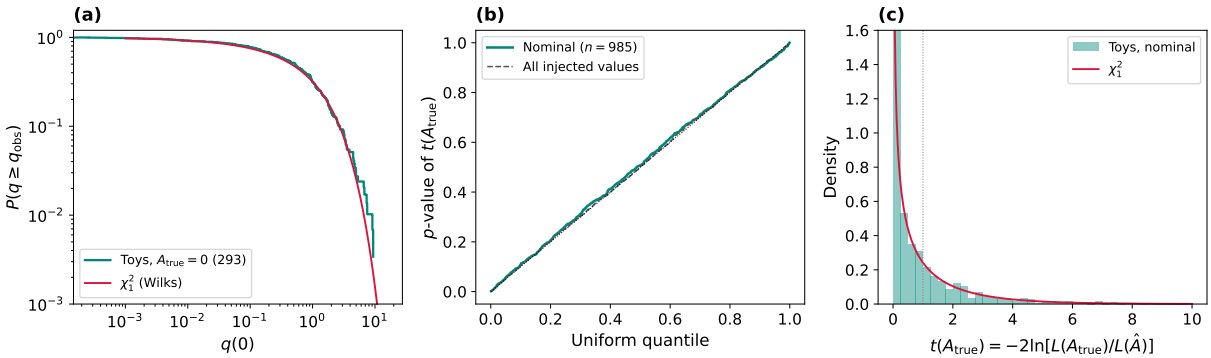


FIG. XXII. Toy-MC experiments, validity of Wilks' theorem. (a) Fraction of experiments with $q(0)$ above the observed value, for $A_{\text{true}} = 0$, compared with the χ^2_1 distribution. (b) p-value of $t(A_{\text{true}})$ against a uniform distribution, at the fitted value and pooled over the injected values. (c) Distribution of $t(A_{\text{true}})$ at the fitted value and χ^2_1.

Appendix F: Validation of the Estimator and of the Intervals

In short: how the toy-MC experiments are generated, the bias, efficiency and coverage of the estimator, its asymptotic behavior, the size and power of the test of $A = 0$ and the goodness of fit as a function of the binning.

1. Generation of the Toy-MC Experiments

In short: toy experiments are generated from the fitted model (Poisson yields, masses by the transformation method) and fitted with the nominal procedure: 1000 *at the fitted value and* 300 *at each of four other values of A.*

2. Bias, Efficiency and Coverage

In short: the bias is at most 0.0008*, the spread equals the Hesse uncertainty within* 6%*, and the coverage at the fitted value is* 0.696 ± 0.015 *and* 0.954 ± 0.007*.*

3. Asymptotic Behavior of the Estimator

In short: with the number of candidates the bias tends to zero, the spread scales as $N^{-1/2}$ and the distribution of $\hat{A}$ becomes Gaussian, as the central limit theorem and the asymptotic properties of the maximum-likelihood estimate predict.

N	bias	sd($\hat{A}$)	sd$\cdot\sqrt{N}$	skewness	ex. kurtosis	Shapiro p
50	−0.0085	0.1611	1.14	−0.15	+0.64	0.03
150	−0.0051	0.0920	1.13	+0.16	+0.31	0.59
500	−0.0022	0.0501	1.12	−0.08	+0.38	0.52
2000	+0.0009	0.0225	1.01	−0.20	−0.20	0.32

TABLE XIII. Distribution of the maximum-likelihood estimate of A in 300 toy-MC datasets of N candidates (six-parameter model of Sec. B.4).

4. Size and Power of the Test of $A = 0$

In short: the test has the nominal size, the statistic $q(0)$ follows a χ^2 distribution with one degree of freedom, and an asymmetry of −0.02 *would be rejected at 3σ in one toy-MC experiment out of four.*

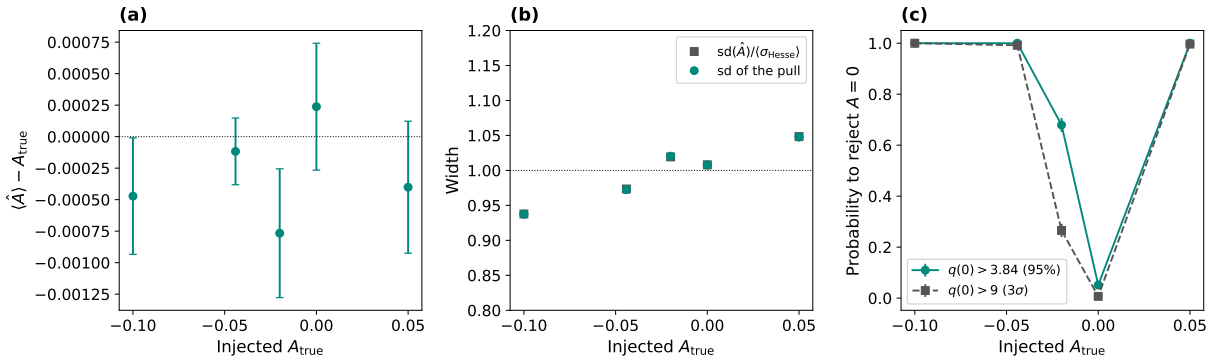


FIG. XXIII. Toy-MC experiments at five injected asymmetries. (a) Bias of $\hat{A}$, with its standard error. (b) Spread of the estimates divided by the mean Hesse uncertainty, and width of the pulls. (c) Probability to reject $A = 0$ in the test with $q(0) > 3.84$ (size 5%) and $q(0) > 9$ (3σ), as a function of the injected value: size at $A_{\text{true}} = 0$, power elsewhere.

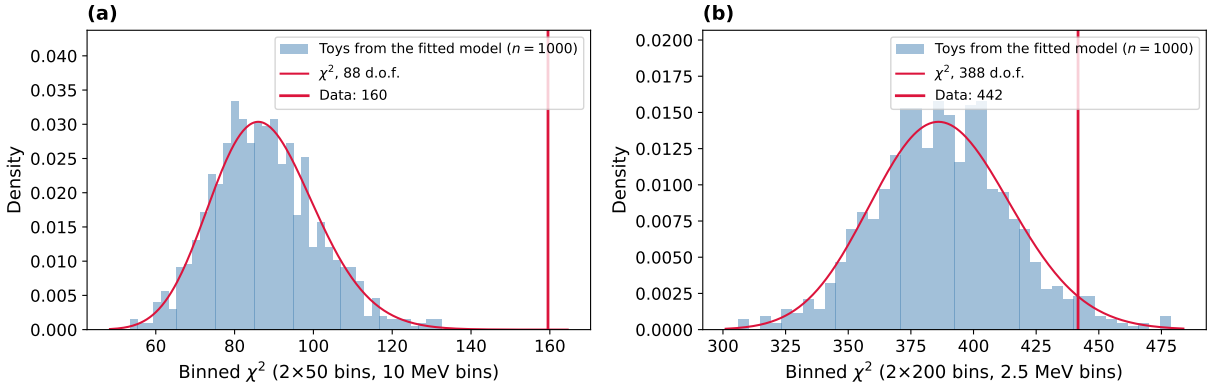


FIG. XXIV. Binned χ^2 of 1000 toy-MC experiments from the fitted model, χ^2 distribution with the corresponding degrees of freedom, and value of the data: (a) 2×50 bins of 10 MeV, (b) 2×200 bins of 2.5 MeV.

5. Goodness of Fit and the Binning

In short: the binned χ^2 of toy-MC experiments follows its asymptotic distribution, so its p-values are accurate; the misfit of the data is 5.4 standard deviations at 10 MeV bins and 2.0 at 2.5 MeV, because it lives on the scale of the flanks of the peak.

Appendix G: Local Asymmetry in the Dalitz Plane

In short: the local deviations and their robustness, the null tests of the procedure, the region of the published analysis and the split by magnet polarity.

1. Local Deviations and Their Robustness

In short: two bins have $|S_i| > 3$ (-3.7 and -3.0) and the global homogeneity test gives $\chi^2 = 80.8$ for 31 degrees of freedom ($p = 3 \times 10^{-6}$), also in the signal window alone.

2. Homogeneity Test: Signal Window, Vetoes, Grids and Magnet Split

In short: the homogeneity test holds in the signal window, with the charm vetoes and for other grids, and the structure is the same in the two magnet polarities ($\chi^2 = 23.0$ for 29 bins).

3. Null Tests of the Local-Asymmetry Procedure

In short: permuting the charges gives $\chi^2 = 31.5 \pm 8.4$*, as expected, and never reaches the observed* 80.8*; the phase-space simulation is consistent with no asymmetry, with slightly optimistic* χ^2 *p-values.*

4. Distribution of the Homogeneity Statistic under Charge Permutation

In short: shuffling the charges reproduces the χ^2 *distribution of a uniform asymmetry, and the observed value lies far beyond all* 2000 *permutations.*

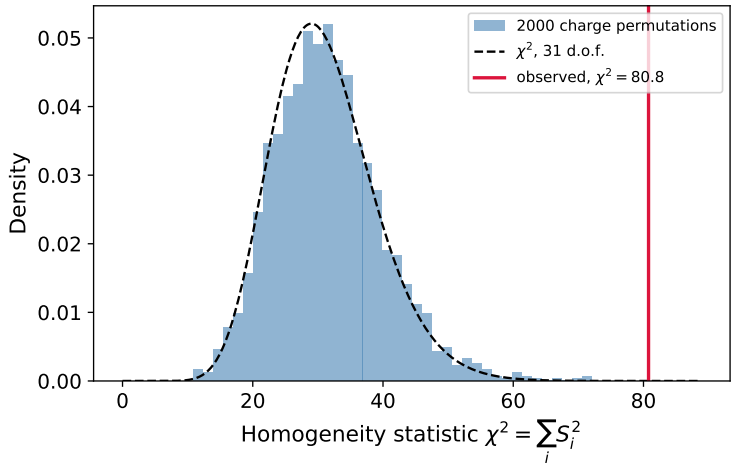


FIG. XXV. Homogeneity statistic $\chi^2 = \sum_i S_i^2$ of the 8×8 Dalitz grid for 2000 random permutations of the charges (histogram), χ^2 distribution with 31 degrees of freedom (dashed) and observed value (line).

5. Asymmetry in the $\phi(1020)$ Band and in the Significant Bins

In short: the band of the $\phi(1020)$ *has no asymmetry; the negative asymmetry is outside it, and larger when the* ϕ *candidates are removed.*

6. Region of the Published Analysis

In short: the $\phi(1020)$ *band has no asymmetry (*-0.004 ± 0.023*); the fit of the* 2146 *candidates of the published region gives* -0.234 ± 0.025 *(Bayesian) and* -0.237 ± 0.025 *(frequentist), against* -0.239 ± 0.020*.*

Appendix H: Comparison of the Two Approaches

In short: the numerical agreement of the two fits, the meaning of the intervals, the hypothesis tests, the profile likelihood against the posterior, and the two methods at low statistics.

1. Numerical Agreement

In short: the two fits agree to the quoted digits (estimates differ by 0.0002*, both uncertainties are* 0.0085*, systematic uncertainties* 0.0070 *and* 0.0072*); sideband subtraction gives* -0.0503 ± 0.0083*.*

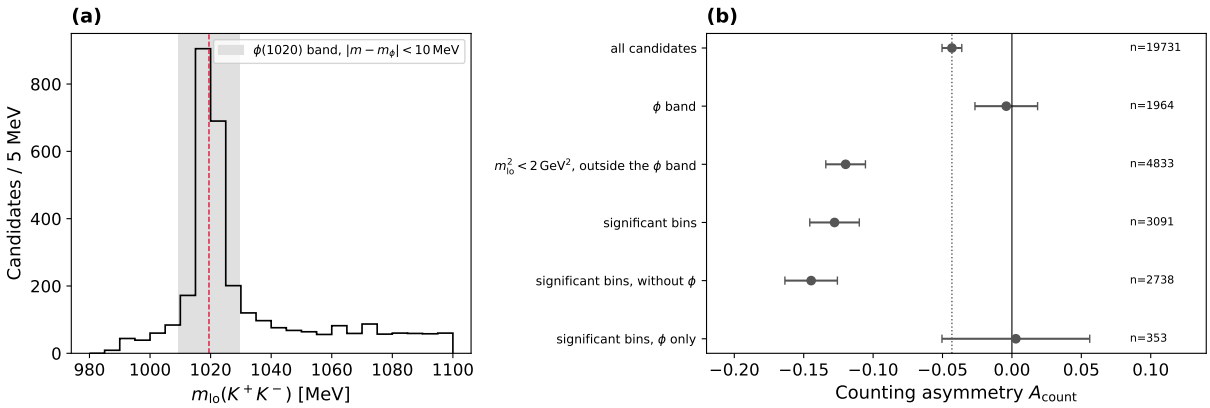


FIG. XXVI. (a) Distribution of $m_{\mathrm{lo}}(K^+K^-)$ around the $\phi(1020)$ (5 MeV bins) and band used (shaded). (b) Counting asymmetry, with binomial uncertainty and number of candidates, of all candidates (dotted line), of the ϕ band, of the region $m^2_{\mathrm{lo}} < 2$ GeV^2 outside the band, and of the two significant bins of Fig. VII, with and without the candidates of the ϕ band, and of those candidates alone.

2. Meaning of the Interval

In short: the credible interval is a probability on A given the data and the prior, the confidence interval a property of the procedure, which the bootstrap and the toy experiments verified.

3. Profile Likelihood, Posterior and the Two Polarities

In short: profiling and marginalizing the nuisance parameters give the same distribution of A; the two magnet polarities agree with each other, and combining their likelihoods reproduces the joint fit.

	Frequentist	Bayesian
Magnet down	-0.0492 ± 0.0111	-0.0497 ± 0.0112
Magnet up	-0.0374 ± 0.0131	-0.0378 ± 0.0129
Combination of the two	-0.0443 ± 0.0084	-0.0446 ± 0.0086
Joint fit	-0.0441 ± 0.0085	-0.0443 ± 0.0085

TABLE XIV. A_{raw} for the two magnet polarities and their combination, by the two methods; the frequentist combination is the sum of the profile likelihoods of the two polarities, the Bayesian one the sequential update of the posterior.

4. Confidence and Credible Regions in (A, A_b)

In short: the 68% and 95% regions of the frequentist profile (ΔNLL at the χ^2_2 values 2.30 and 5.99) and of the Bayesian highest posterior density coincide, and $A = 0$ lies outside both.

5. Hypothesis Test

In short: the exact binomial test on all candidates gives 6.1σ, the test of the fit $p = 2 \times 10^{-7}$ (5.2σ) and the Bayes factor $BF_{10} \approx 9 \times 10^3$; they answer different questions.

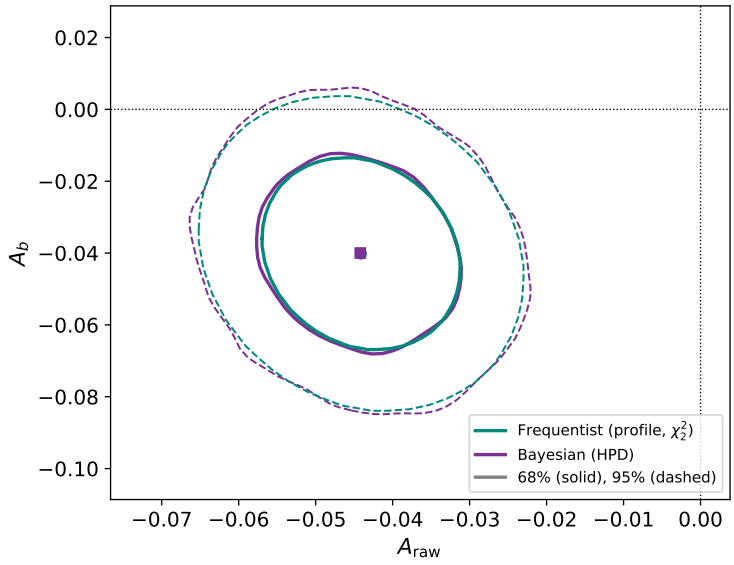


FIG. XXVII. 68% (solid) and 95% (dashed) regions in (A, A_b): frequentist, from the profile likelihood with χ^2_2 thresholds, and Bayesian, highest-posterior-density regions of the chain. Dots: best-fit and posterior median.

6. Goodness of Fit

In short: the same fit has $p = 0.03$ *in* 2.5 *MeV bins and* $p \approx 5 \times 10^{-6}$ *in* 10 *MeV bins: the p-value of the binned* χ^2 *is a property of the binning as much as of the model.*

7. Nuisance Parameters and Validation

In short: the frequentist analysis profiles the eleven nuisance parameters and the Bayesian one marginalizes them; the results coincide because A *is almost uncorrelated with them.*

8. Published Analysis: Samples and Models

In short: the published analysis has $22{,}119 \pm 164$ *signal candidates, a Cruijff signal and misidentified and partially reconstructed backgrounds; with the same* D^0 *veto the Bayesian fit gives* -0.0496 ± 0.0088.

9. Published Analysis: Raw versus CP Asymmetry

In short: $A_{CP} \approx A_{\text{raw}} - A_P - A_D$*; the corrections move the published result from* -0.060 *to* $-0.043 \pm 0.009 \pm 0.003 \pm 0.007$ *and cannot be assumed for a different selection.*

10. Machine-Learning Cross-Check

In short: a cut at 0.5 *on the classifier score keeps* 92% *of the signal and* 36% *of the background and gives* -0.0451 ± 0.0087 *(Bayesian) and* -0.0447 ± 0.0086 *(frequentist).*

11. Bayesian and Frequentist Intervals at Low Statistics

In short: credible and confidence intervals coincide for $\sim 2 \times 10^4$ *candidates with flat priors, and stay equal, in value and coverage, down to* 50 *candidates.*

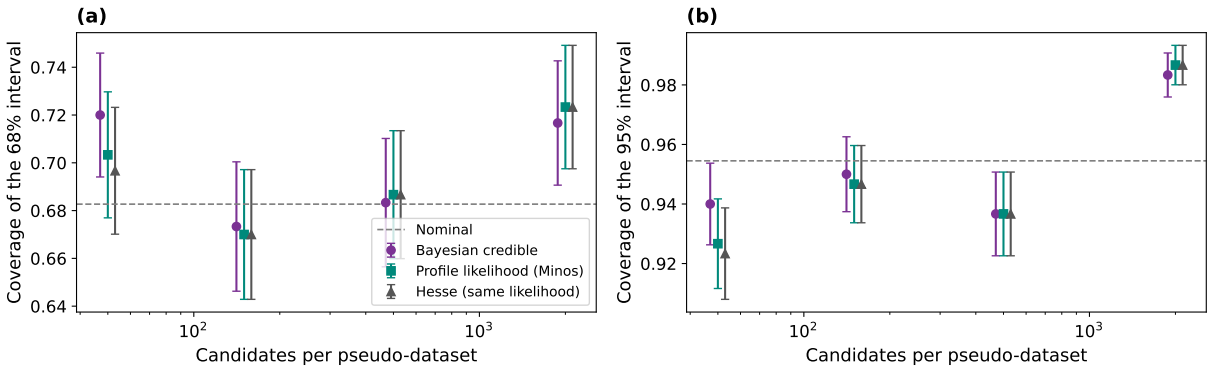


FIG. XXVIII. Coverage of the (a) 68% and (b) 95% intervals on A vs the number of candidates, for the Bayesian credible interval, the profile-likelihood (Minos) interval and the Hesse interval of the same likelihood; 300 toy-MC datasets per point, binomial uncertainties. The dashed lines mark the nominal coverage (1σ, 2σ; the Bayesian 95% interval is the 2.5–97.5% one).

Appendix I: Machine-Learning Analysis

In short: classifiers trained on weak labels from the mass spectrum (there are no true labels) reach a test AUC of 0.86*; a cut on their score leaves* A_{raw} *unchanged, and an adaptive partition of the Dalitz plane shows the same pattern as the fixed binning.*

1. Background: Regularization, Bias-Variance and Loss Functions

In short: cross-entropy is a Bernoulli negative log-likelihood, $L1$ *and* $L2$ *penalties shrink the coefficients, the test error splits into bias, variance and noise, and class weights rescale the score by a known factor.*

2. Background: Classification Without Labels

In short: the classifier that is optimal between two mixtures of the same signal and background is monotonic in the signal-to-background ratio, so labels derived from the mass are sufficient.

3. Background: Clustering and Uncertainty Estimators

In short: K*-means converges to a local minimum of the inertia, the adjusted Rand index measures the stability of a partition, and the spread of an ensemble estimates the epistemic uncertainty.*

4. Weak Labels and Feature Set

*In short: a classifier separates a signal-enriched window from the mass sidebands (*15,130 *signal-like and* 1940 *background-like candidates,* 20 *features,* 60/20/20 *split); by the Neyman-Pearson argument of the CWoLa approach it is also the signal/background classifier, if the features carry no information on the mass.*

5. Model Comparison

In short: six classifiers; the tree ensembles and the neural network reach test AUC 0.857–0.864*, about* 0.09 *above the linear models and indistinguishable from each other within the test-set uncertainty (*0.012*); gradient boosting, best in cross-validation, is the reference.*

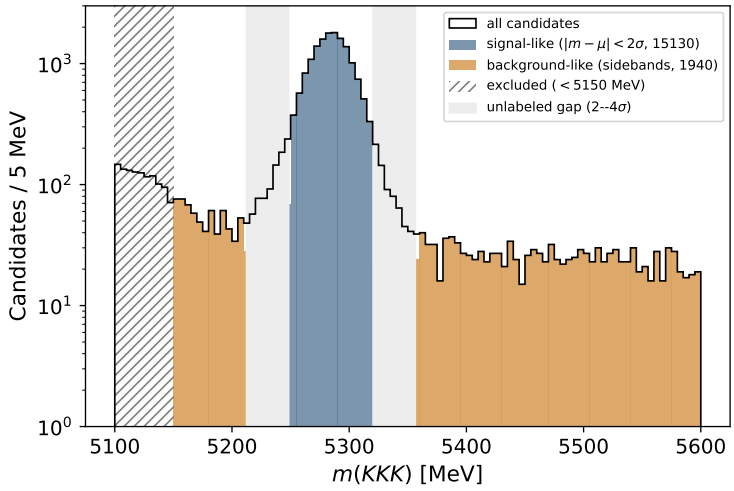


FIG. XXIX. Weak labels on the mass spectrum of the fit window: signal-like candidates ($|m-\mu| < 2\sigma$, 15,130) and background-like candidates (sidebands beyond 4σ and above 5150 MeV, 1940); the 2–4σ gap is left unlabeled and the region below 5150 MeV, where the background rises towards partially reconstructed decays, is excluded.

Model	CV AUC	Test AUC	Brier
Logistic Regression	0.769 ± 0.016	0.771 ± 0.013	0.197
Logistic Regression (L1)	–	0.771 ± 0.013	0.198
Decision Tree	0.816 ± 0.011	0.824 ± 0.013	0.133
Random Forest	0.863 ± 0.017	0.860 ± 0.012	0.122
Gradient Boosting	$\mathbf{0.866 \pm 0.016}$	0.864 ± 0.012	0.116
FFNN	–	0.857 ± 0.012	0.134

TABLE XV. Cross-validated AUC (mean and standard deviation over the folds), test-set AUC (with bootstrap uncertainty) and test-set Brier score of the raw scores. Gradient boosting, best in cross-validation, is the reference model. The FFNN is validated on the validation set instead of by cross-validation.

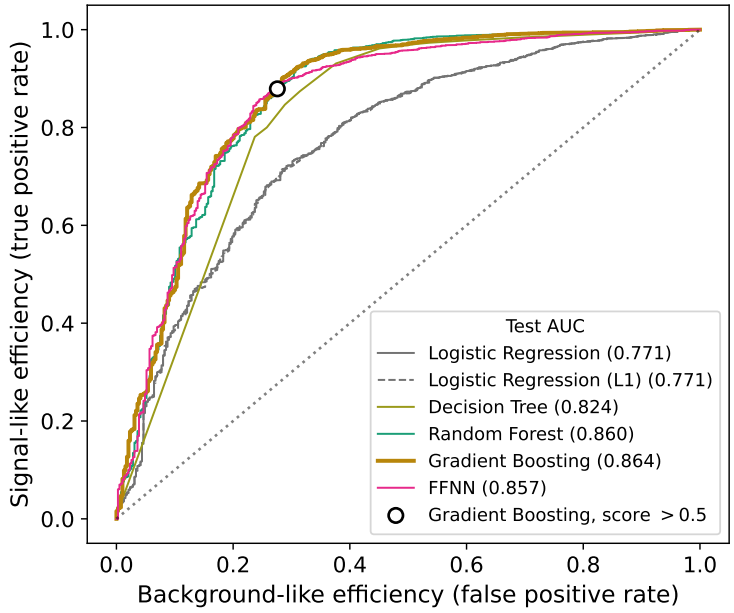


FIG. XXX. ROC curves on the test set, with the test AUC of each model in the legend (Table XV); the reference model is drawn thicker, and the two logistic regressions overlap. The circle marks the cut on the reference score used in Appendix I.9.

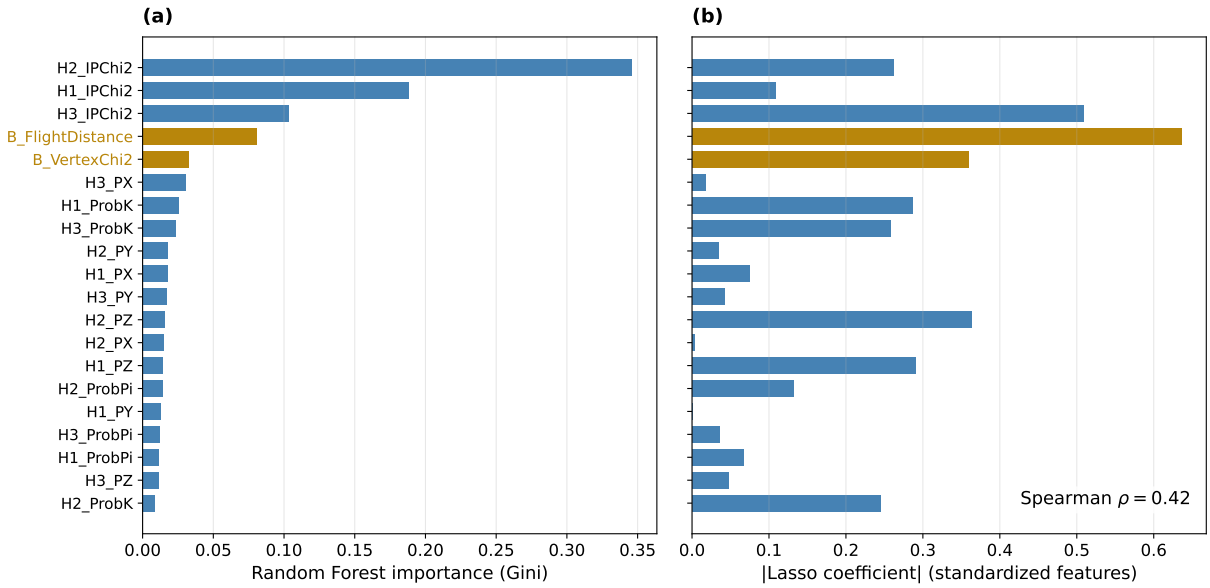


FIG. XXXI. Random Forest feature importance (Gini, a) and magnitude of the Lasso coefficient of the standardized feature (b) for the 20 features, both ordered by Random Forest importance; the two B-candidate variables are highlighted.

6. Regularization, Bias-Variance, and Calibration

In short: the linear model does not overfit, gradient boosting overfits beyond ∼150 stages without losing test AUC, and class weights bias the probabilities, which an isotonic recalibration corrects (Brier score $0.122 \to 0.062$*) without changing the ranking.*

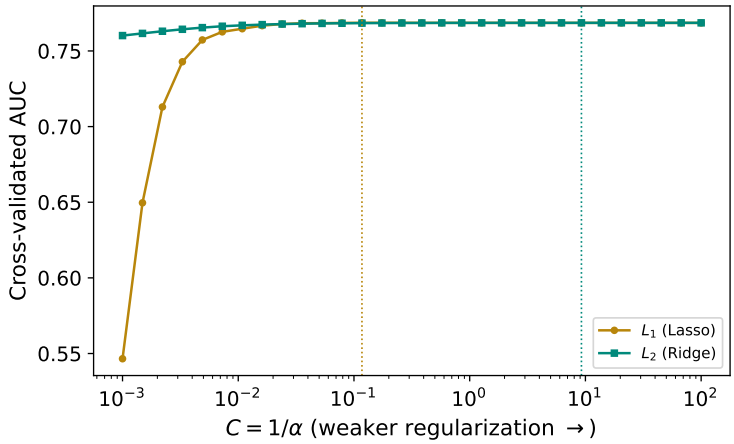


FIG. XXXII. Cross-validated AUC of the logistic regression against the inverse regularization strength $C = 1/\alpha$ (30 log-spaced values), for the L_1 (Lasso) and L_2 (Ridge) penalties; the dotted lines mark the optimum of each.

7. Uncertainty Quantification

In short: the three estimators give mean per-candidate uncertainties of 0.178 *(Random Forest),* 0.095 *(MC dropout) and* 0.050 *(deep ensemble); they measure different sources of variability, and only the Random Forest spread visibly decreases with the training-set size.*

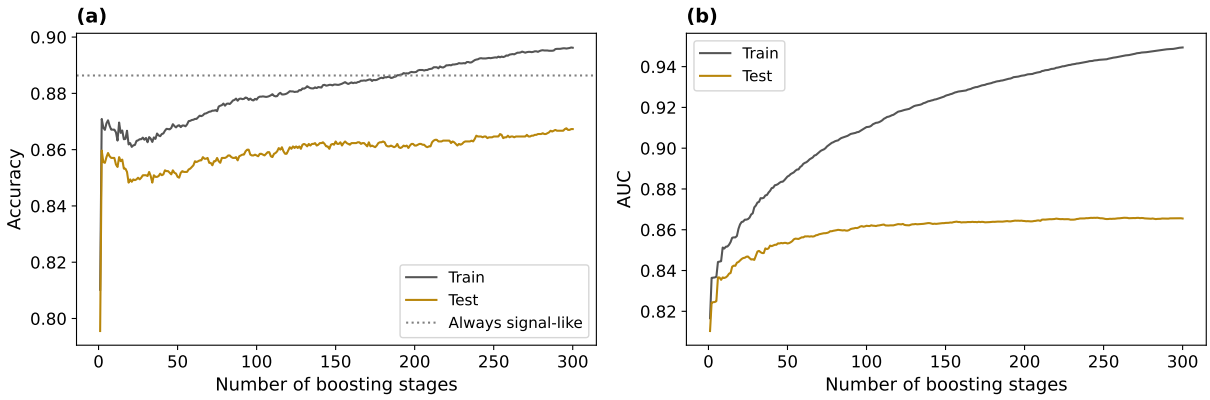


FIG. XXXIII. Gradient boosting: (a) accuracy at the 0.5 threshold and (b) AUC on the training and test sets as a function of the number of boosting stages. The dotted line is the accuracy of always predicting the signal-like class.

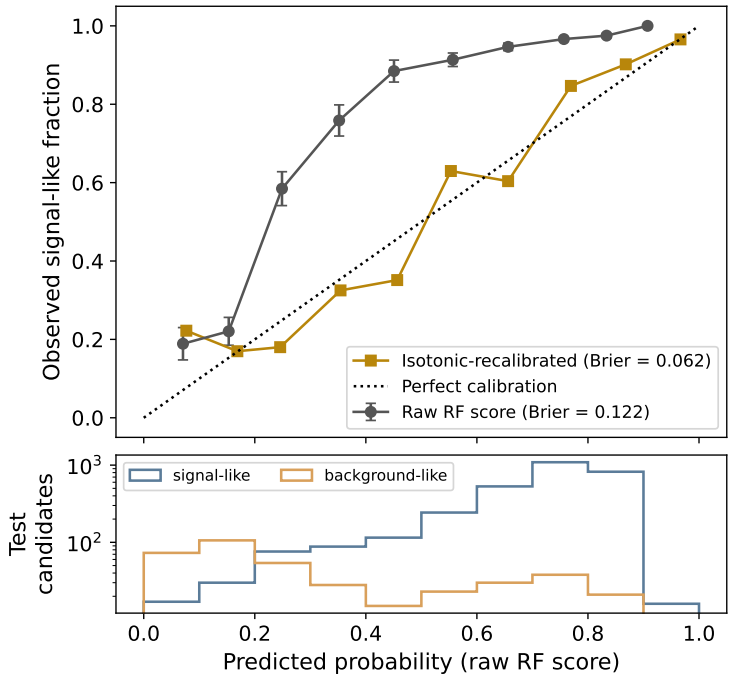


FIG. XXXIV. Reliability diagram of the Random Forest on the test set: observed fraction of signal-like candidates vs. predicted probability, for the raw score (with binomial uncertainties) and for the isotonic-recalibrated one. Bottom: distribution of the raw score for the two weak-label classes; the low-probability bins contain few candidates.

8. K-means on the Dalitz Plot

In short: for $K = 8$ two clusters deviate by more than 3σ ($\chi^2 = 46.4$ for 7 degrees of freedom): a negative asymmetry at low $m^2_{\rm lo}$ and positive ones at high $m^2_{\rm hi}$, as in the fixed binning.

9. Stability of $A_{\rm raw}$ under ML Selection

In short: a cut on the score at 0.5 keeps 92% of the signal and 36% of the background and leaves $A_{\rm raw}$ within its expected statistical spread with both fits and with counting; it does not improve the precision and cannot remove the misidentified background.

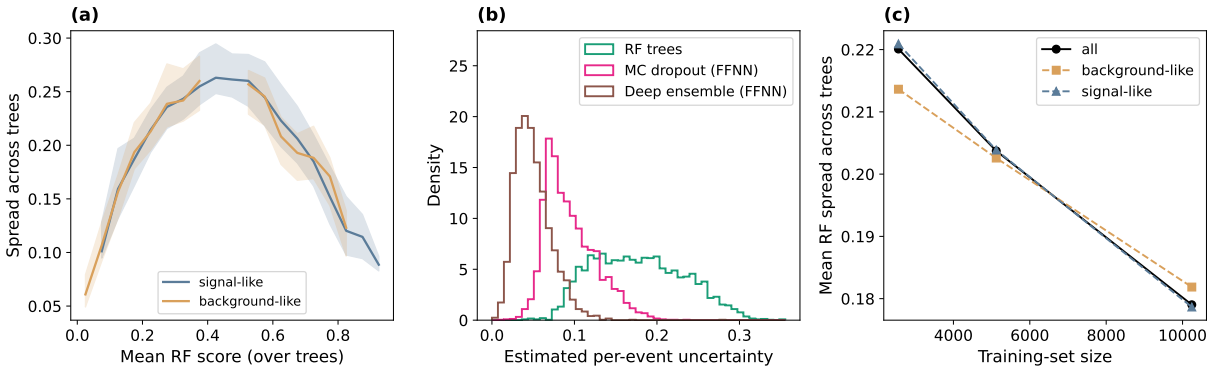


FIG. XXXV. (a) Spread across the Random Forest trees vs. mean tree score, median and 16–84% band of the test candidates of each weak-label class. (b) Distributions of the per-candidate uncertainty from the Random Forest trees, MC dropout and the deep ensemble. (c) Mean Random Forest spread vs. training-set size, for all, background-like and signal-like test candidates.

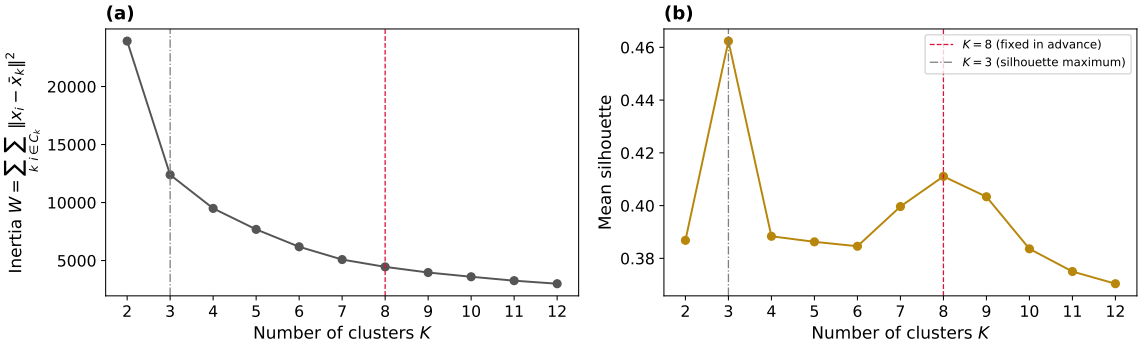


FIG. XXXVI. (a) Inertia and (b) mean silhouette of the K-means partition of the Dalitz plane for $K = 2$–12; the dashed line marks the $K = 8$ fixed in advance and the dash-dotted line the maximum of the silhouette, $K = 3$.

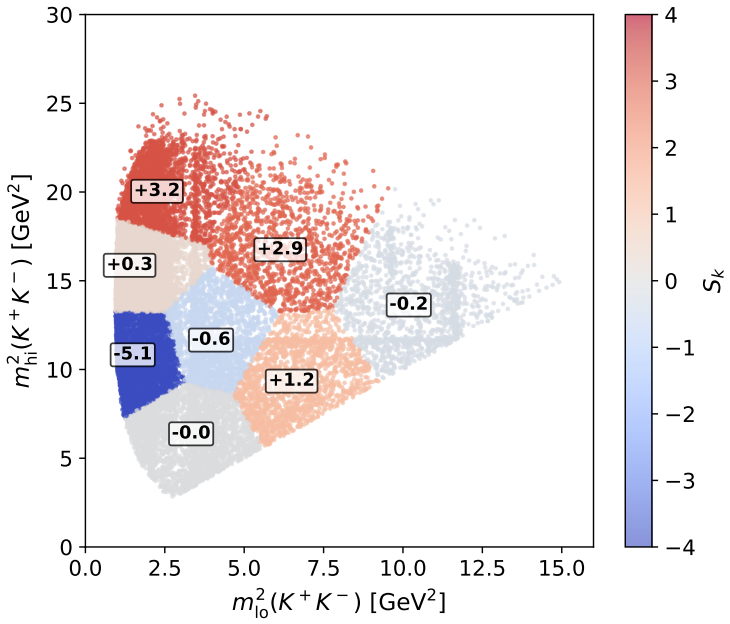


FIG. XXXVII. K-means partition ($K = 8$) of the Dalitz plane, all candidates of the fit window, same axes as Fig. VII. Color and labels: local significance of each cluster with respect to the global asymmetry, $S_k = (A_k - A_{\rm glob})/\sigma(A_k)$, on the same scale as panel (c) of Fig. VII.

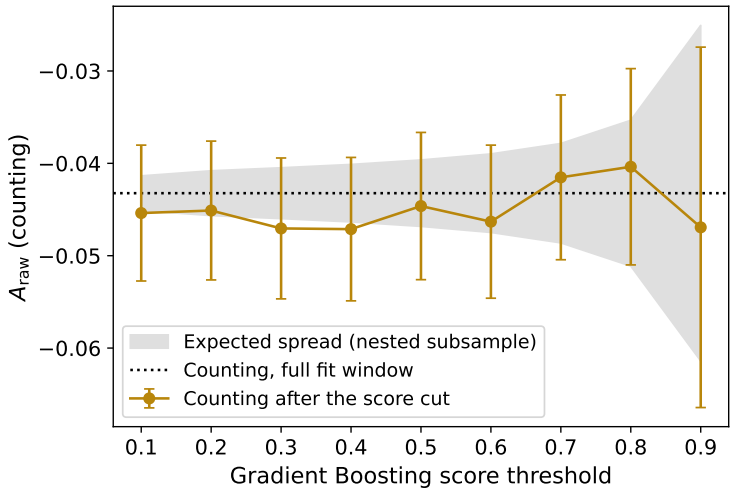


FIG. XXXVIII. Counting asymmetry after a cut on the gradient-boosting score, vs. threshold, with its binomial uncertainty. Dotted line: counting asymmetry of the full fit window; band: expected statistical spread of a nested subsample around it, $\pm\sqrt{\sigma^2_{\rm sub} - \sigma^2_{\rm full}}$.

Method	Full sample	Score > 0.5	Shift (expected)
Bayesian fit	-0.0443 ± 0.0085	-0.0451 ± 0.0087	$-0.0008\,(0.0018)$
Frequentist fit	-0.0441 ± 0.0085	-0.0447 ± 0.0086	$-0.0006\,(0.0012)$
Counting	-0.0432 ± 0.0071	-0.0446 ± 0.0080	$-0.0014\,(0.0036)$

TABLE XVI. $A_{\rm raw}$ on the full fit window (19,731 candidates) and after the cut on the gradient-boosting score (15,709), by the three methods. The expected shift is the statistical spread of a nested subsample, $\sqrt{\sigma^2_{\rm sub} - \sigma^2_{\rm full}}$. Counting includes the background.

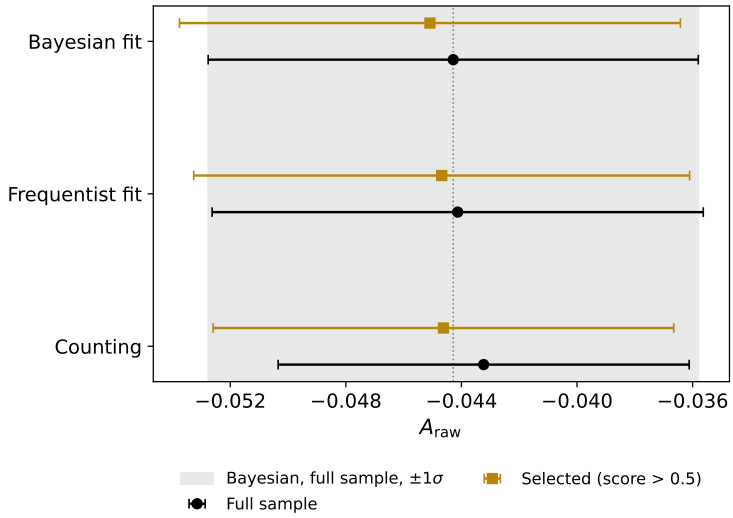


FIG. XXXIX. $A_{\rm raw}$ from the Bayesian fit, the frequentist fit and counting, for the full sample and the ML-selected one (Table XVI).

10. Hyperparameter Optimization

In short: the best configurations improve the test AUC by at most 0.005*, less than its uncertainty, and the cross-validated maximum overestimates the test AUC (winner's curse).*

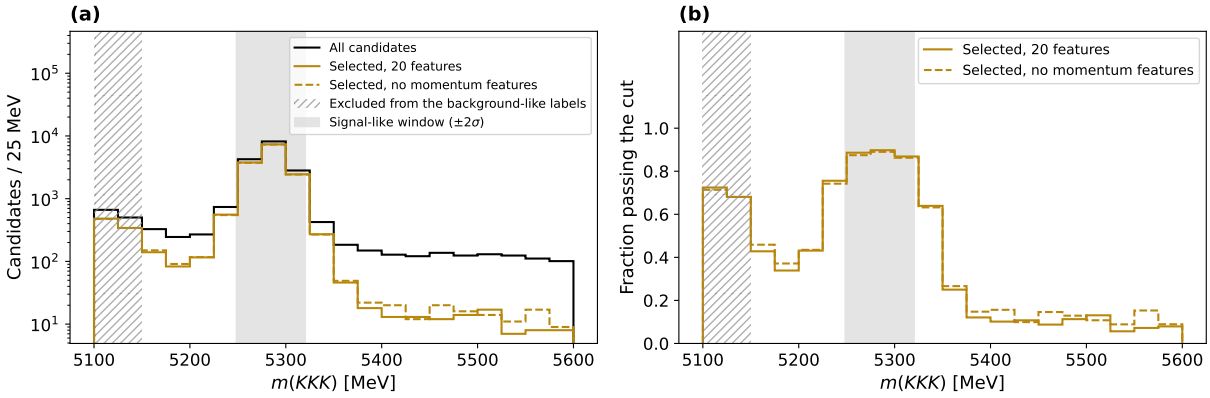


FIG. XL. (a) Mass spectrum of all candidates and of those with gradient-boosting score > 0.5, for the 20-feature classifier and for one trained without the kaon momentum components (25 MeV bins). (b) Fraction of candidates passing the cut. Hatched: the region below 5150 MeV excluded from the background-like labels; grey: the signal-like window.

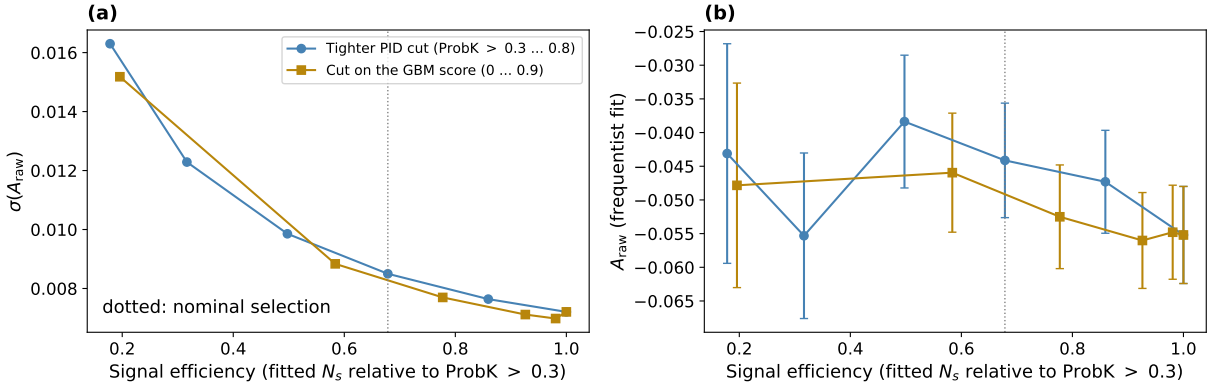


FIG. XLI. High-background sample (`ProbK` > 0.3): (a) statistical uncertainty and (b) value of A_{raw} from the frequentist fit vs signal efficiency, for tighter identification cuts (`ProbK` $> 0.3, \ldots, 0.8$) and for cuts on the score of the classifier retrained on this sample $(0, \ldots, 0.9)$. The dotted line marks the nominal selection (`ProbK` > 0.5). The samples are nested, so their uncertainties are correlated.

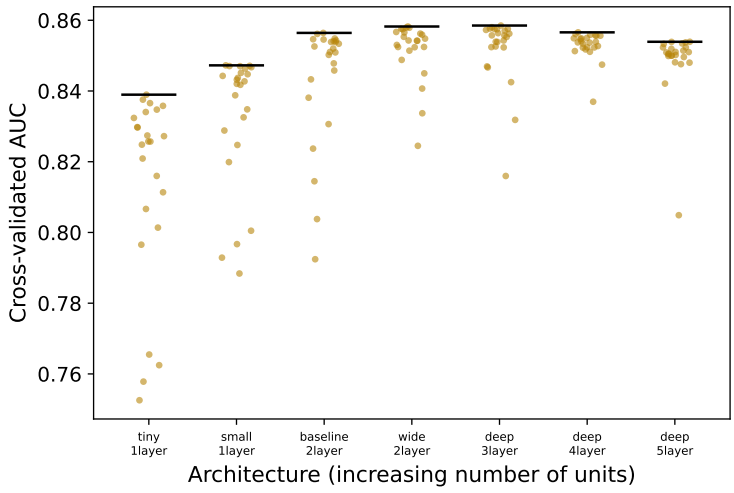


FIG. XLII. Cross-validated AUC of the 168 FFNN configurations (dots) for each of the seven architectures, ordered by number of units; the black segment marks the best configuration of each architecture.

Model	Grid	Best configuration	CV AUC / Test AUC (default)
Random Forest	`n_estimators` 200–1200, `max_depth` 6–12 or none, `min_samples_leaf` 5–40	400, 12, 5	0.8706 / 0.8656 (0.8604)
Gradient Boosting	`n_estimators` 100–500, `max_depth` 1–5, `learning_rate` 0.01–0.1	300, 2, 0.1	0.8684 / 0.8650 (0.8643)

TABLE XVII. Hyperparameter search for the tree ensembles (120 and 100 configurations, 5-fold cross-validation repeated 10 times, balanced class weights). The test AUC of the best configuration is compared with that of the default of Table XV (in parentheses); the bootstrap uncertainty of each test AUC is 0.012.

Architecture	Layers	Dropout	Learning rate	CV AUC
`deep_3layer` (best)	$[64, 32, 16]$	0.2	0.003	0.8585 ± 0.0174
`wide_2layer`	$[64, 32]$	0.3	0.003	0.8582 ± 0.0175
`deep_3layer`	$[64, 32, 16]$	0.3	0.003	0.8580 ± 0.0187
`wide_2layer`	$[64, 32]$	0.3	0.005	0.8579 ± 0.0184
`wide_2layer`	$[64, 32]$	0.4	0.005	0.8579 ± 0.0187

TABLE XVIII. Top five of the 168 FFNN configurations (7 architectures from one hidden layer of 8 units to five layers $[128, 64, 32, 16, 8]$, dropout 0.1–0.4, learning rates 0.0002–0.005; 5-fold cross-validation repeated twice). The five are within 0.0007 of each other, far below the fold-to-fold spread: the optimum is a plateau. The best configuration has test AUC 0.8606, against 0.8568 for the default architecture (± 0.005 over initializations).

11. Machine Learning and the Fits

In short: the machine-learning selection is a cross-check: a cut at 0.5 *leaves* $A_{\rm raw}$ *within its expected spread with both methods, does not improve the precision, and cannot remove the misidentified decays.*

12. Local Asymmetries: Fixed Binning and K-Means

In short: the fixed grid and the K*-means partition show the same pattern, a negative asymmetry at low* $m^2_{\rm lo}$ *and positive ones at high* $m^2_{\rm hi}$ *(*$\chi^2 = 80.8/31$ *and* $46.4/7$*).*

Appendix J: Conclusions, Scope and Limitations

In short: the scope of the study and the channels of the record that are not analysed, the complete text of the conclusions, and the limitations of the result.

1. Scope of the Study and Channels of the Record

In short: the aim is a methodological comparison of the two frameworks on the same data, not a measurement of A_{CP}*, which this record cannot give for lack of the control channel; the other three channels are not analysed.*

2. Headline Result

In short: $A_{\rm raw} = -0.0441 \pm 0.0085 \pm 0.0072$ *(frequentist) and* $-0.0443 \pm 0.0085 \pm 0.0070$ *(Bayesian),* 5.2 *standard deviations from zero, stable under the thirteen variants; it is a raw asymmetry, not a significance of CP violation.*

3. Lessons of Each Method

In short: the frequentist fit gives intervals with nominal coverage and an unbiased estimator; the Bayesian one the same interval with flat priors plus diagnostics to check, a Bayes factor that depends on the prior and a predictive check; the ML selection leaves $A_{\rm raw}$ *unchanged.*

4. Systematic Study and Dependence on Choices

In short: the particle-identification requirement dominates, and several statistics depend on choices: the binned χ^2 on the binning, the Bayes factor on the prior, the coverage on how the toy experiments are generated.

5. Limitations and Future Work

In short: no signal model reaches $\chi^2/\mathrm{ndof} \approx 1$ although A_{raw} does not depend on it, the machine-learning labels are weak and imbalanced, A_{raw} is not corrected for production and detection asymmetries, and the natural next step is a control-channel correction.

Appendix K: Code and Data Availability

In short: Python code with the package versions listed here; frequentist, Bayesian and machine-learning code and notebooks are in the accompanying repository. All code is written in Python ≥3.12, with `scikit-learn` [14] for the classical machine-learning models and `TensorFlow/Keras` [15] for the neural networks. Key package versions: `uproot 5.7`, `iminuit 2.32.0`, `emcee 3.1.6`, `corner 2.3.0`, `scikit-learn 1.9`, `tensorflow 2.21.0`, `keras 3.15.1`, `scipy 1.18`, `numpy 2.4`, `pandas 3.0`.

The dataset is the CERN Open Data record 4900 ("Matter Antimatter Differences – B meson decays to three hadrons"). The analysis code is available in the accompanying repository, in `Code_general/Frequentist/` (frequentist fits, the common systematic study, and `freq_validation.py` with the toy-MC experiments, profile likelihoods, confidence belt and bootstrap studies of Sec. VII and Appendix F), `Code_general/Bayesian/` (notebooks `01_bayes_mass_fit_and_asymmetry`: nominal fit, Bayes factors, systematic study, toy-MC closure test, magnet polarities and Dalitz plane; `02_bayes_model_checks_and_priors`: posterior predictive checks, convergence diagnostics, sensitivity to the priors and comparison with the profile likelihood; `03_bayes_vs_frequentist_low_statistics`: the two approaches at low statistics; the mass model, the sampler and the toy generator are in `bayes_model.py`), and `Code_general/Machine_Learning/` (notebooks `04_ml_signal_background_classifiers`: weak labels, classifiers, calibration, uncertainty and hyperparameters; `05_ml_selection_and_local_asymmetry`: effect of the selection on A_{raw}, K-means and null tests on the Dalitz plane, comparison with the published analysis; helper functions in `ml_tools.py`). Each of these notebooks opens with a table of the sections and figures of this paper that it produces and generates those figures from its saved results, so that they can be rebuilt without repeating the long computations; the common style is in `Code_general/plot_style.py`. The figures of the frequentist analysis are produced from its saved results by `regen_figures_validation.py`, `regen_figures_systematics.py` and `regen_figures_extras.py` in the figures directory of the report. All random numbers are generated with fixed seeds, so the toy-MC experiments can be reproduced.

Appendix L: Notation

In short: the symbols and abbreviations used in the report.

[1] R. Aaij *et al.* (LHCb), Direct CP violation in charmless three-body decays of $B^{\pm}$ mesons, Phys. Rev. D **108**, 012008 (2023), arXiv:2206.07622 [hep-ex].

[2] L. Falcão, CP violation measurements in three-body charmless B decays, on behalf of the LHCb Collaboration, 12th Workshop on the CKM Unitarity Triangle (2024), arXiv:2403.19921 [hep-ex].

[3] LHCb Collaboration, Matter Antimatter Differences – B meson decays to three hadrons (2017), record 4900.

[4] LHCb Collaboration, Matter Antimatter Differences – Project Notebook (2017), record 4902.

[5] LHCb Collaboration, R. Aaij, *et al.*, Measurement of CP violation in the phase space of $B^{\pm} \to K^{\pm}\pi^{+}\pi^{-}$ and $B^{\pm} \to K^{\pm}K^{+}K^{-}$ decays, Physical Review Letters **111**, 101801 (2013), arXiv:1306.1246.

[6] H. Dembinski, P. Ongmongkolkul, *et al.*, scikit-hep/iminuit, Zenodo (2020).

[7] F. James and M. Roos, Minuit – a system for function minimization and analysis of the parameter errors and correlations, Computer Physics Communications **10**, 343 (1975).

[8] S. S. Wilks, The large-sample distribution of the likelihood ratio for testing composite hypotheses, Annals of Mathematical Statistics **9**, 60 (1938).

[9] J. Goodman and J. Weare, Ensemble samplers with affine invariance, Commun. Appl. Math. Comput. Sci. **5**, 65

Symbol	Meaning
$A_{\rm raw}$	raw charge asymmetry $(N^- - N^+)/(N^- + N^+)$ of the signal, parameter A of the fit
A_{CP}, A_P, A_D	CP asymmetry; production asymmetry of $B^\pm$; detection asymmetry
$A_{\rm count}$	counting asymmetry of all candidates of a sample
A_b	asymmetry of the background yields
N_s, N_b	signal and background yields (both charges)
μ, σ	mean and width of the Gaussian signal
$\lambda^\pm_{1,2}$, $f_\pm$	slopes and fraction of the double-exponential background, per charge
θ	vector of the twelve parameters
$\mathcal{L}$, π	likelihood and prior
$q(A) = 2\Delta{\rm NLL}$	profile likelihood ratio of A; $t(A_{\rm true})$ at the injected value
Z, p	significance and p-value of the test of $A = 0$
BF_{01}, BF_{10}	Bayes factor of $A = 0$ against A free, and its inverse
p_B	posterior predictive p-value
σ_B	expected spread of the shift between nested samples, $\sqrt{\lvert\sigma^2_{\rm var} - \sigma^2_{\rm nom}\rvert}$
N/τ	chain length in integrated autocorrelation times
S_i, S_k	local significance of a Dalitz bin or cluster with respect to the global asymmetry

TABLE XIX. Main symbols used in the text.

(2010).

[10] D. Foreman-Mackey, D. W. Hogg, D. Lang, and J. Goodman, emcee: The MCMC Hammer, Publ. Astron. Soc. Pac. **125**, 306 (2013), arXiv:1202.3665.

[11] J. M. Dickey, The weighted likelihood ratio, linear hypotheses on normal location parameters, Ann. Math. Statist. **42**, 204 (1971).

[12] H. Jeffreys, *Theory of Probability*, 3rd ed. (Oxford University Press, Oxford, 1961).

[13] R. Barlow, Systematic errors: facts and fictions, in *Conference on Advanced Statistical Techniques in Particle Physics* (Durham, 2002) arXiv:hep-ex/0207026.

[14] F. Pedregosa *et al.*, Scikit-learn: Machine learning in Python, Journal of Machine Learning Research **12**, 2825 (2011).

[15] M. Abadi *et al.*, TensorFlow: Large-scale machine learning on heterogeneous distributed systems (2016), arXiv:1603.04467.